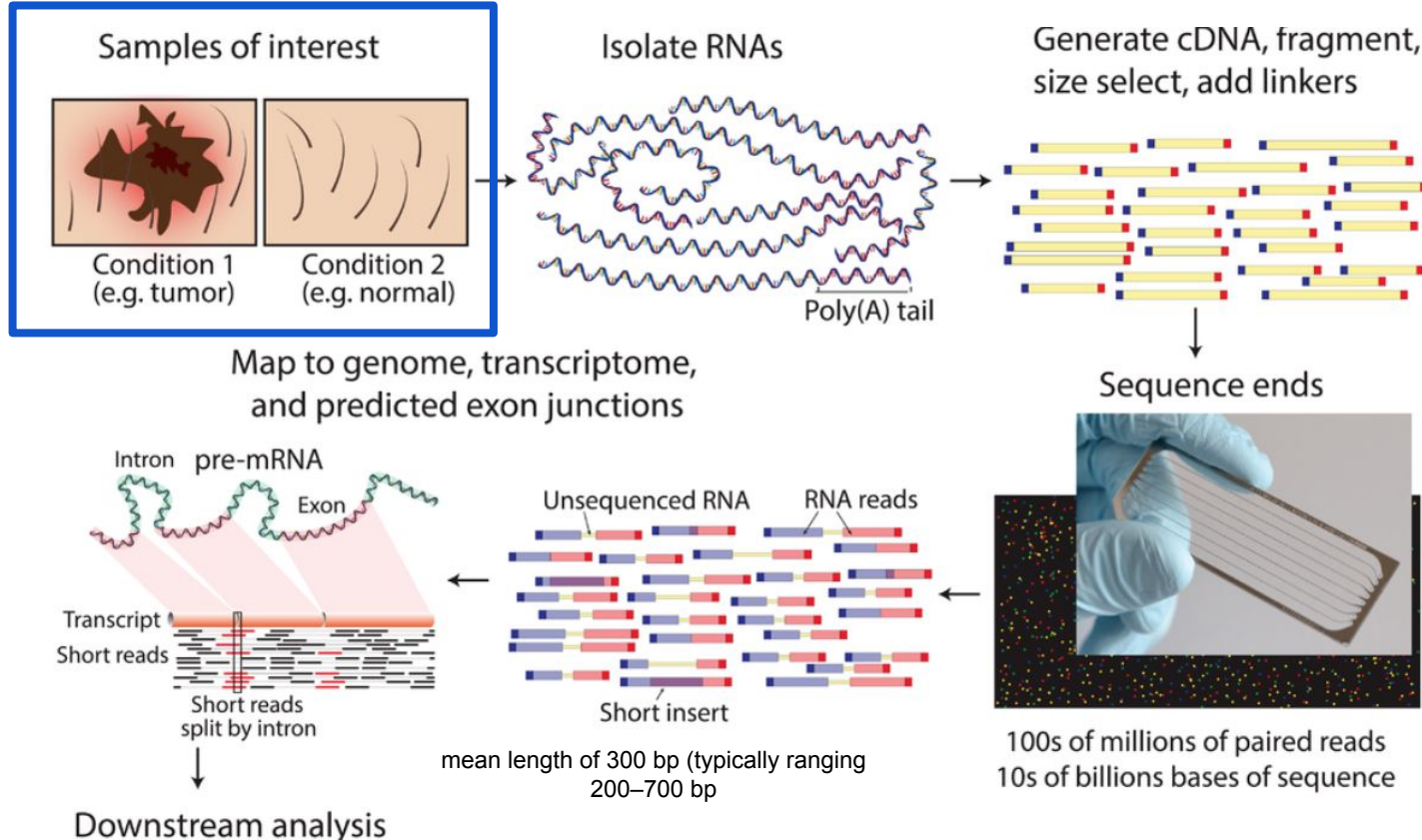


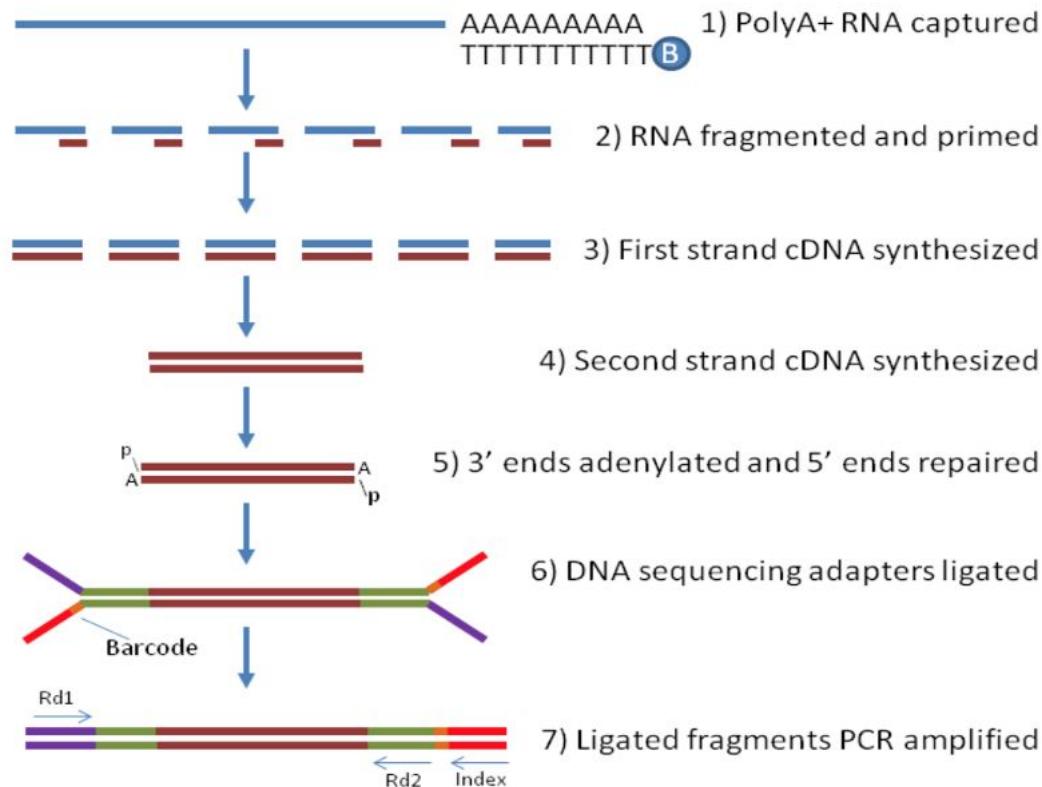
Library preparation

Please revise slides 2-8 before the workshop

RNA-seq - experiment



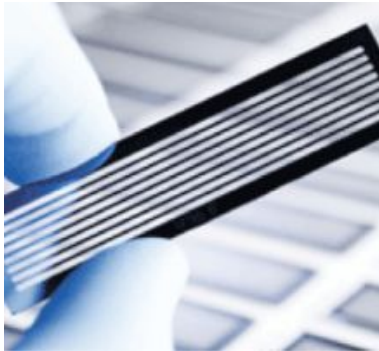
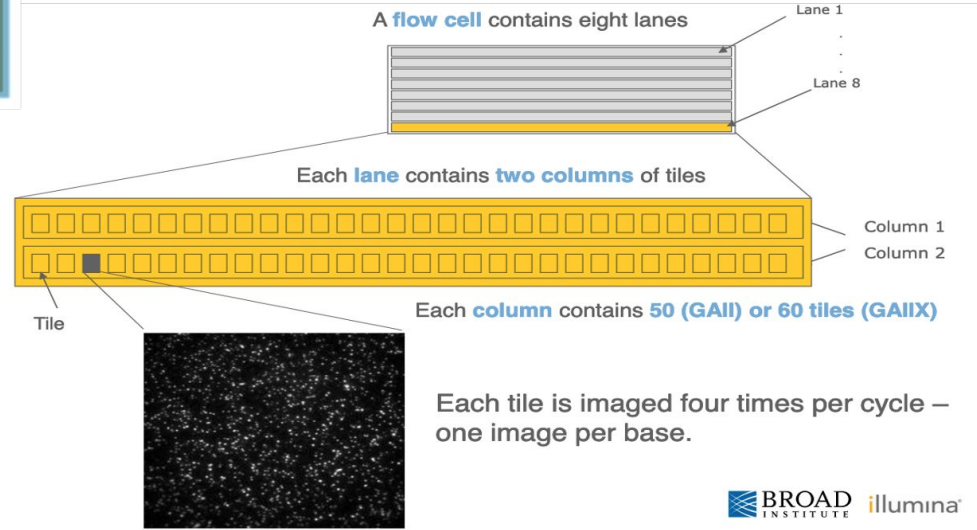
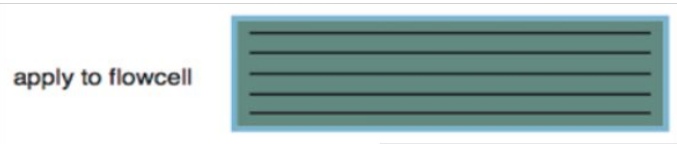
How does the bulk-RNA-seq technology work?



Flow cell organization for sequencing



cDNA library is applied to a flow cell



Fragments preparation for the sequencer

Single index



Unique dual index



xGen UDI-UMI adapter



Flow cell binding sequence: Platform-specific sequences for library binding to instrument



Sequencing primer sites: Binding sites for general sequencing primers



Sample indexes: Short sequences specific to a given sample library

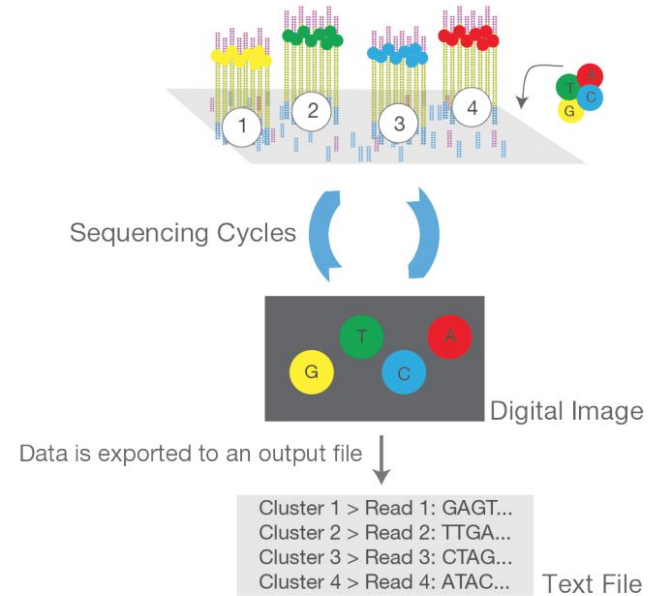
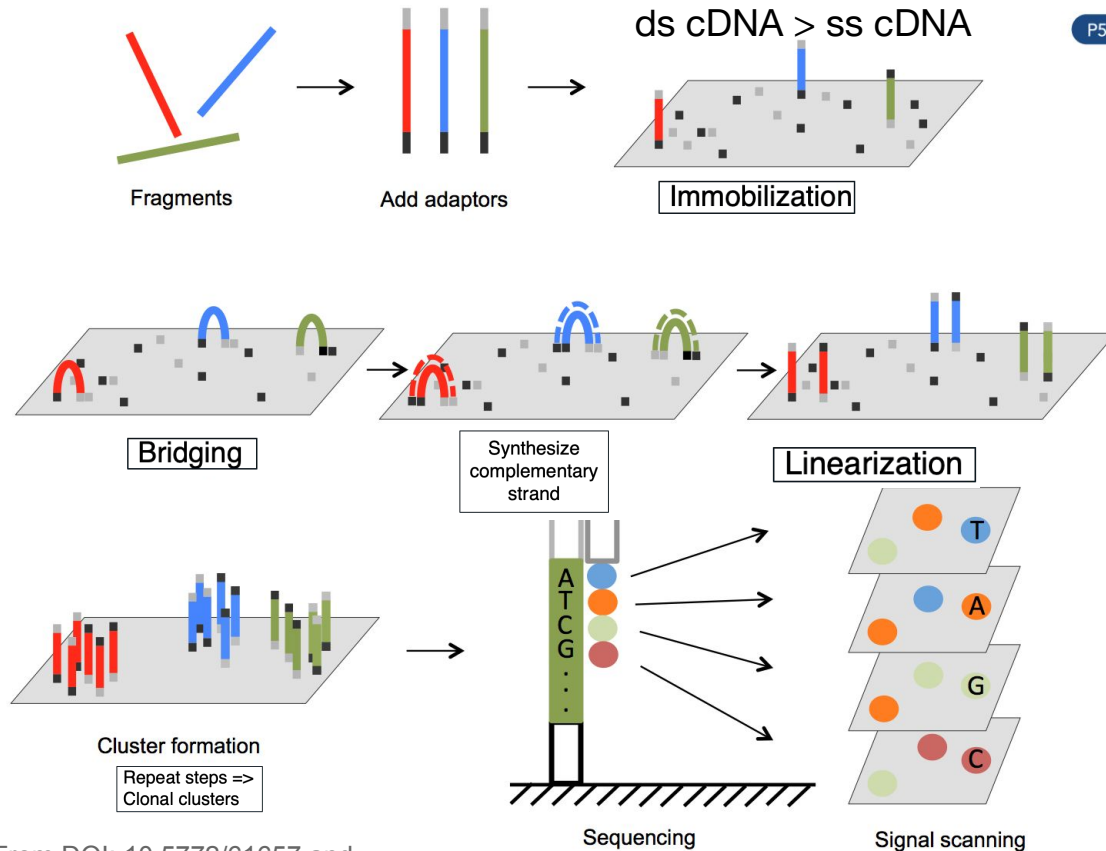


Molecular index/barcode: Short sequence used to uniquely tag each molecule in a given sample library



Insert: Target DNA or RNA fragment from a given sample library

DNA fragments immobilized on flow cell & amplified into clonal clusters



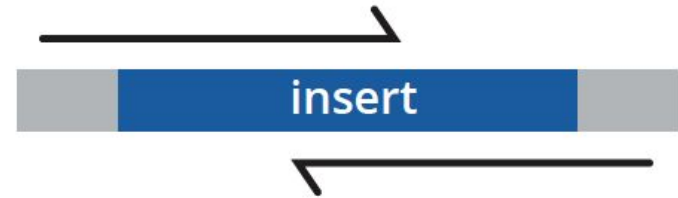
Types of sequencing

Single-end sequencing



- unidirectionally - sequencing from one end only
- less expensive, is typically reserved for quantifying gene expression in well-annotated genomes and analyzing small RNA molecules.

Paired-end sequencing



- bidirectionally - both ends of DNA fragments to be sequenced
- usually recommended for most applications as it provides richer data and permits longer library insert sizes.

Library preparation affects the downstream analysis

- RNA quality is fundamental
- RNA extraction method can affect the transcript abundance
- Choice of platform and library preparation protocol

The background features a dark teal color with several wavy, overlapping lines in a lighter teal shade. These lines are composed of a grid of small, dashed segments, creating a sense of depth and movement. The lines flow from the left side towards the right, with some peaking and others dipping, resembling a stylized landscape or a data visualization.

GLADSTONE INSTITUTES

Introduction to RNA-seq data analysis

Michela Traglia, Ayushi Agrawal
Bioinformatics Core, GIDB
September 28-29, 2026

GLADSTONE
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Gladstone Bioinformatics Workshops

All training materials are free to use by anyone at any time. Instructors regularly present these workshops at Gladstone and the UCSF Mission Bay Campus. Follow Gladstone's [Data Science Training Program](#) for upcoming events.

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Learning Paths

Explore these curated sets of workshops to find the right path for you. Note that intermediate and advanced workshops require completion of earlier workshops before enrolling.

Biostats	Command Line	R Scripting
		
Statistics for biologists	Computer skills for data science	Data analysis using R

RNA-Seq	ATAC-seq	Network Biology	Pathway Analysis
			
RNA-Seq analysis	scATAC-seq analysis	Network analysis and visualization	Pathway analysis and modeling

- Also check out the workshops offered by [UCSF's Data Science Initiative](#).



Scan me!

RNA-Seq Analysis

 NOVICE LEVEL

[Introduction to RNA-Seq Analysis](#)

 INTERMEDIATE LEVEL

[Intermediate RNA-Seq Analysis Using R](#)

 ADVANCED LEVEL

[Single Cell RNA-Seq Analysis](#)

 ADVANCED LEVEL

[Advanced RNA-seq Analysis Using R](#)

 INTERMEDIATE LEVEL

[Introduction to RNA-Seq Analysis](#)



Introductions

Michela Traglia

Senior Statistician

Ayushi Agrawal

Senior Bioinformatician

Workshop sessions

Session I

Monday - September 28
9-12p

Session II

Monday - September 28
1-3 pm

Session III

Tuesday - September 29
1-3 pm

Working material for this workshop

Session I-II-III

1. Slides
2. **Session I-II** - Hands-on materials
3. **Session III** - Hands-on materials

Please install Docker

<https://www.docker.com/products/docker-desktop/>

What to expect from this workshop

- Demystify each step of the RNA-Seq data analysis
- Understand bioinformatic analysis steps from raw data to DEG
- Enable informed conversations with computational biologists
- Demonstrate how to process a mock dataset and get a count matrix using *docker*
- Perform the normalization and DEG analysis with a real dataset using R packages
- Not to provide end-to-end fits all analysis pipeline

Target audience

Session I and II: from sequencer to count matrix

- Ideal for beginners with no prior experience; it might feel slow if you are already familiar with the concepts.

Session III: from count matrix to DEG

- Familiarity with basic concepts of statistics and hypothesis testing

For the demos - expected familiarity with command line, R and RStudio

Poll 1

Which is your expectation from this workshop?
I want to:

- 1) Analyze my RNA-seq raw data
- 2) Improve my existing analysis pipeline settings
- 3) Learn about RNA-seq analysis

Please select all that apply.

Workshop outline

Session 1

From sequencer to adapter trimming

Session 2

From trimming to count matrix

Session 3

From count matrix to DEGs

Workshop outline

Session I

- RNA-seq analysis overview
- Understanding the sequencer output
- Quality checks
- Trimming of adapters

Break

- Introduction to docker and setup
- Demo - QC steps

Session II

- Mapping the read to reference genome
- From alignment to counting features

Break

- Demo - from Mapping to the count matrix

Session III

- Filtering low counts genes
- Dimensionality reduction
- Normalization

Break

- Model fitting
- DEG and visualization

Break

- Demo - from count matrix to DEG

Production-ready pipelines: nf-core/rnaseq

What you will learn today:

Individual tools (FastQC, cutadapt, STAR, featureCounts)

What we use at Gladstone Bioinformatics Core:

Automated pipelines (nf-core/rnaseq)

- Runs all steps automatically
- Handles batch processing (100+ samples)
- Built-in best practices
- Comprehensive quality reports

 [nf-core/rnaseq documentation:
https://nf-co.re/rnaseq](https://nf-co.re/rnaseq)

Why learn manual steps first?

- **Debugging:** Know where the pipeline failed
- **Customization:** Not all projects fit defaults
- **Understanding:** Interpret results properly
- **Troubleshooting:** Fix issues when they arise

Understanding the steps makes you a better bioinformatician and computational biologist.

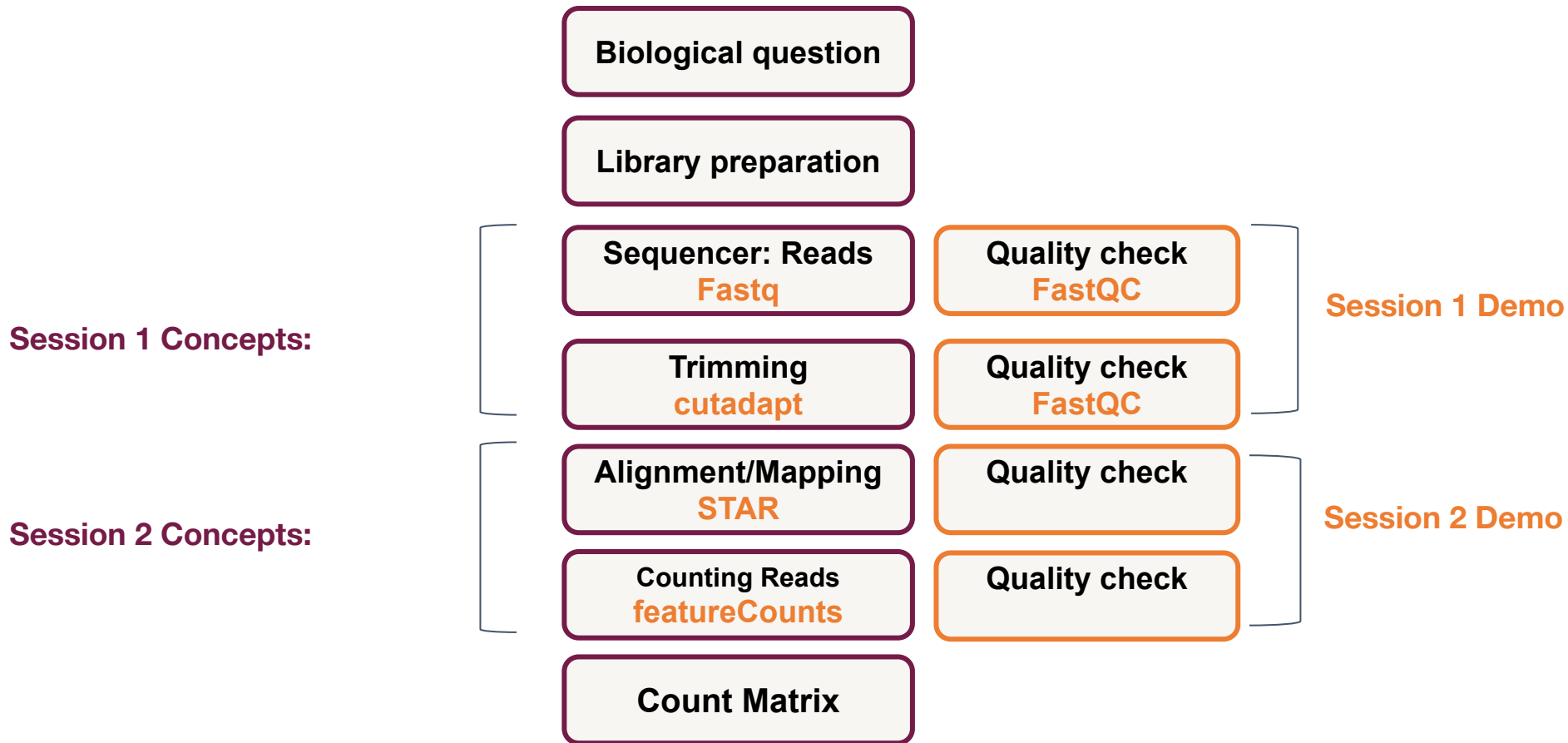
 [nf-core/rnaseq documentation:
https://nf-co.re/rnaseq](https://nf-co.re/rnaseq)

Session 1

From sequencer to adapter trimming

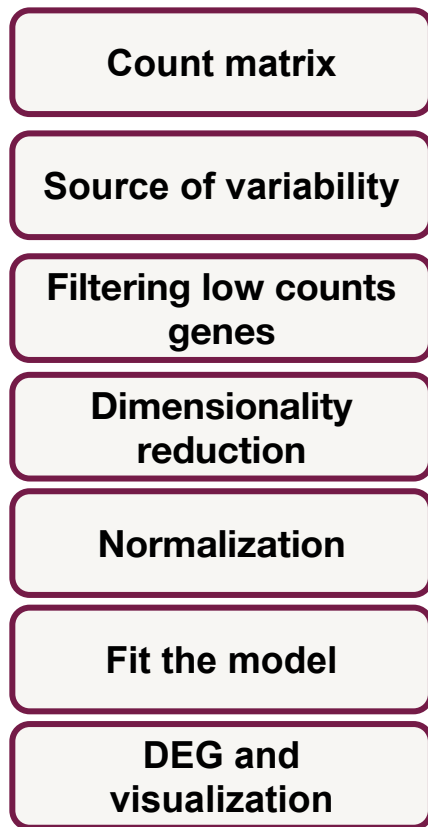
Tools: Docker, FastQC, cutadapt

RNA-seq - analysis workflow (A)



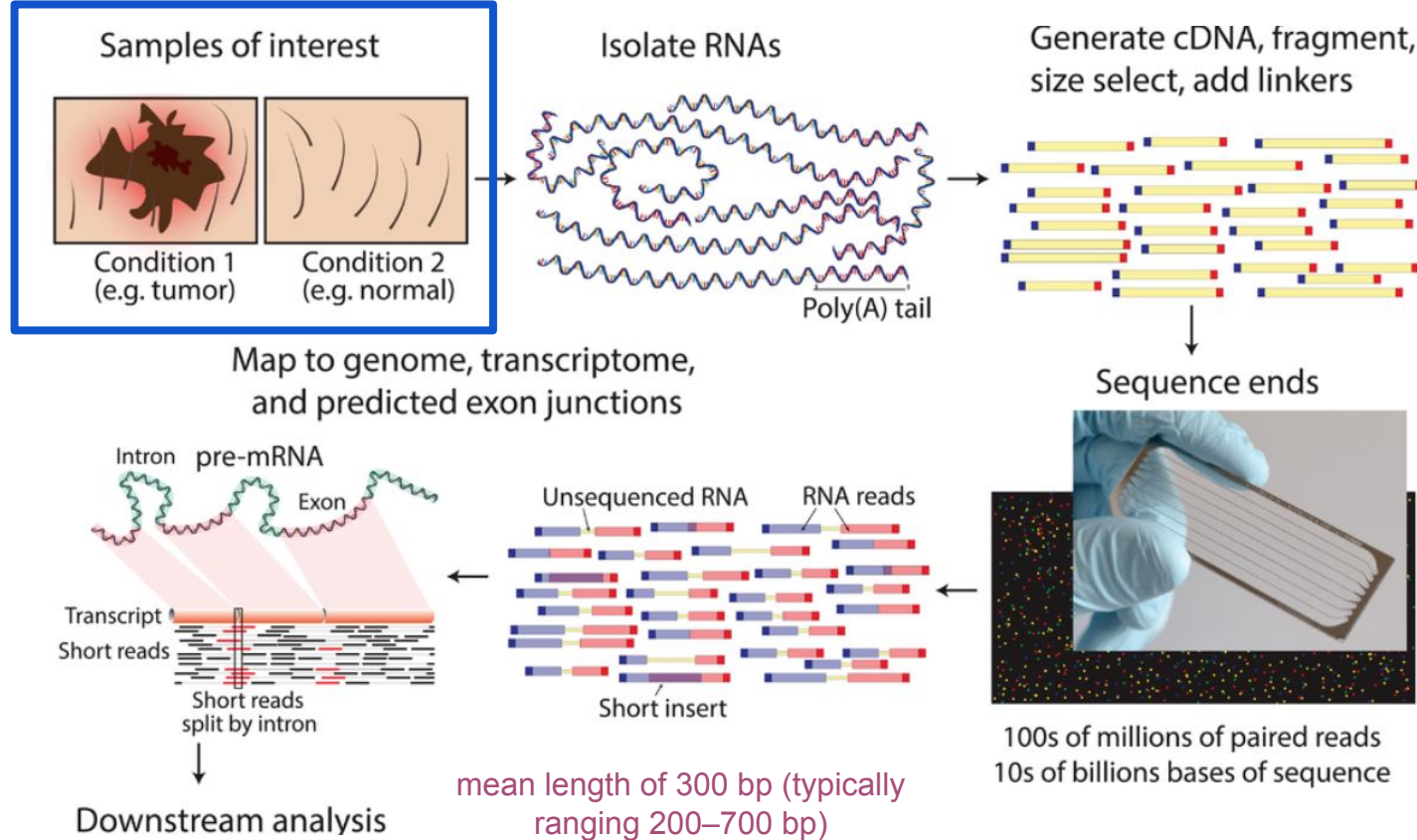
RNA-seq - analysis workflow (B)

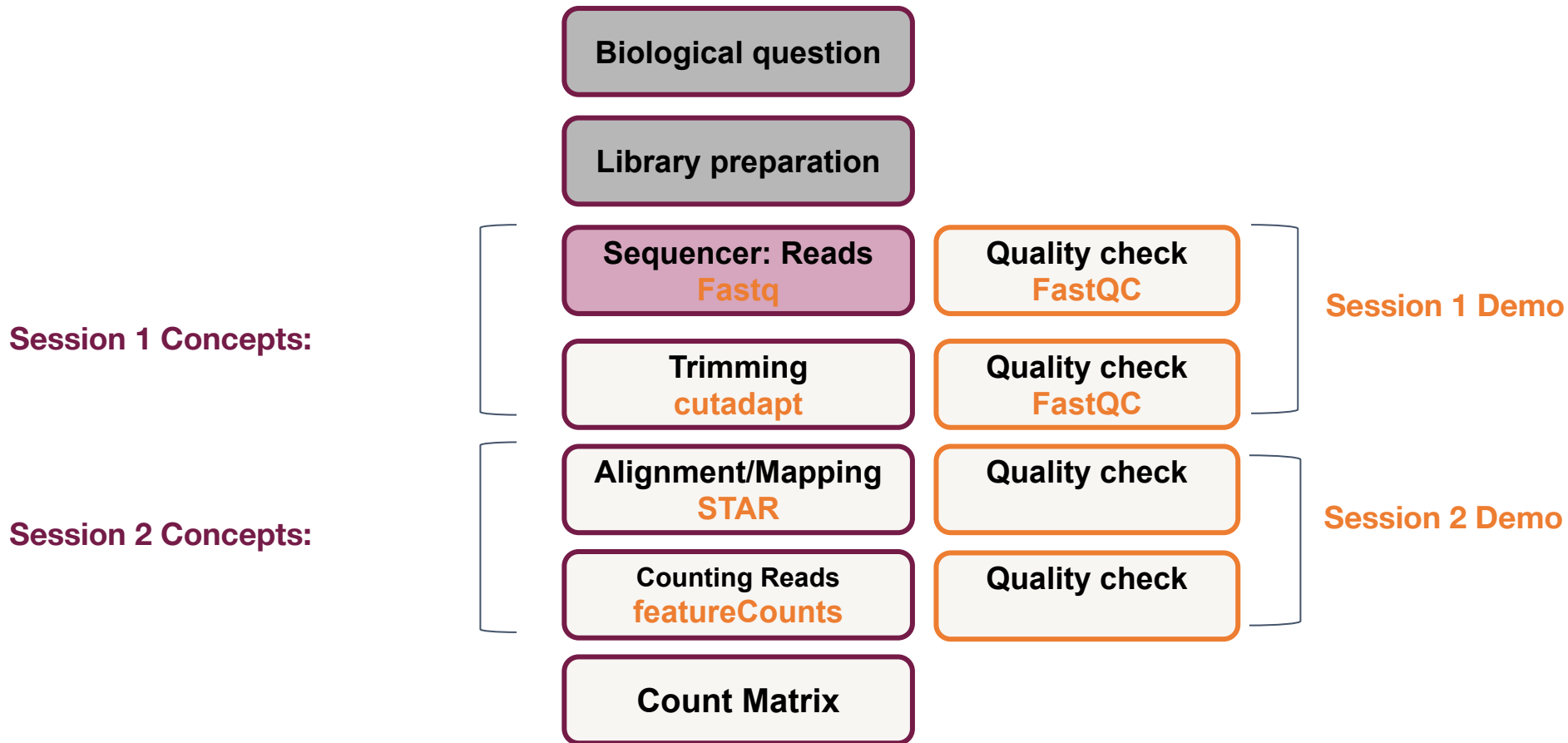
Session 3 Concepts:



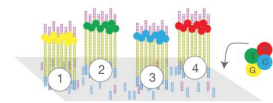
Session 3 Demo

Bulk RNA-seq - experiment





Naming conventions for fastq files



Sequencing Cycles



Digital Image

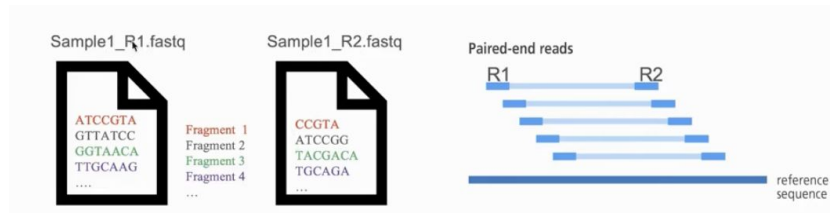
Data is exported to an output file

Cluster 1 > Read 1: GAGT...
Cluster 2 > Read 2: TTGA...
Cluster 3 > Read 3: CTAG...
Cluster 4 > Read 4: ATAC...

Text File

- File names often follow a format.
 - SampleName_SampleNumber_LaneNumber_ReadNumber_SetNumber.fastq
 - Eg – Bacteria_S1_L001_R1_001.fastq
- Paired-end reads named with R1 and R2 in file name.
 - Eg – Bacteria_GATTACA_L001_R1_001.fastq and Bacteria_GATTACA_L001_R2_001.fastq

- File extensions may be .fq or even .txt
- Often compressed using gzip.
 - gzip is free and open-source.
 - Resulting file names have .gz added. Example – .fq.gz.



FASTQ files are text files with detailed information about each read.

The diagram illustrates the structure of a FASTQ record. It consists of four lines of text. The first line is the **Label**, which includes sample and read identifiers. The second line is the **Sequence**, a string of nucleotide bases. The third line is a plus sign. The fourth line contains **Q scores (as ASCII chars)**, which are ASCII values representing the quality of each base. A callout box explains that for a base 'T', the quality score '!' corresponds to a Phred score of 25. Arrows point from the labels to their respective parts in the FASTQ record.

```
@A00564:60:HHJKFDMXX:1:1101:2031:1031 1:N:0:TGGCTTCA+CAACCACA  
CGGACTGGTGGTATGCTGAGTACGTCCCAAGGGTATGGCTGTTCGCCATA  
+  
;;>@:=;=::@B;>A=<=<=B?;;=@@?<?=B;A@=B?>=B:=B=@<<B:;
```

Label

Sequence

Q scores (as ASCII chars)

Base=T, Q='!'=25

Quality score encoding based on ASCII table

Measurement of the likelihood that a base call in DNA sequencing is incorrect

!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/	0	1	2	3	4
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

5	6	7	8	9	:	;	<	=	>	?	@	A	B	C	D	E	F	G	H	I
20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40

Each symbol is a probability p that the base call is incorrect.

The standard Sanger sequencing score to assess reliability of a base call is $Q = -10 \log_{10}(p)$

p , the probability that a given base is incorrectly called

Phred quality score calculation

$$Q = -10 \cdot \log_{10}(P_{\text{err}})$$

Error probability (P_{err})	$\log_{10}(P_{\text{err}})$	Phred quality score	
1	0	0	
0.1	-1	10	
0.01	-2	20	Higher quality scores are better (≥ 20 is considered "good")
0.001	-3	30	
0.0001	-4	40	

A quality score of 20 (Q20) represents an error rate of 1 in 100 (meaning every 100 bp sequencing read may contain an error), with a corresponding call accuracy of 99%.

Poll 2

What is the Q-score (ASCII) of the 3rd base?

@SIM:1:FCX:1:15:6329:1045 1:N:0:2

TCGCACTCAACGCCCTGCATA

+

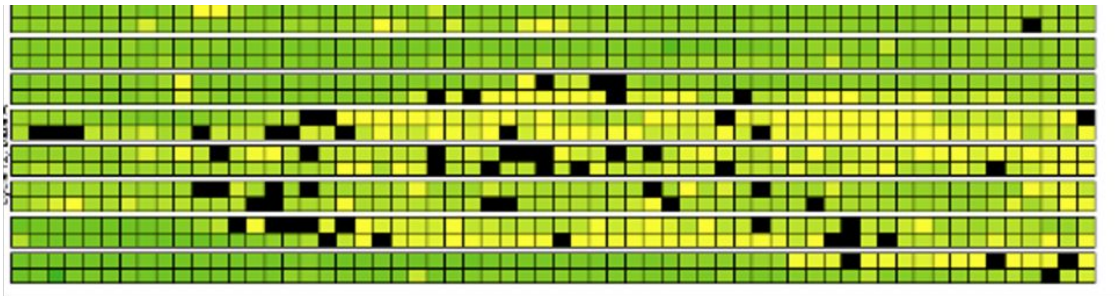
<>;##=><9=AAAAAAAAAA9#

1. G
2. >
3. ;
4. |
5. None of the above

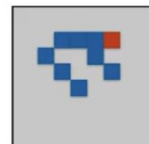
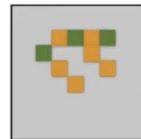
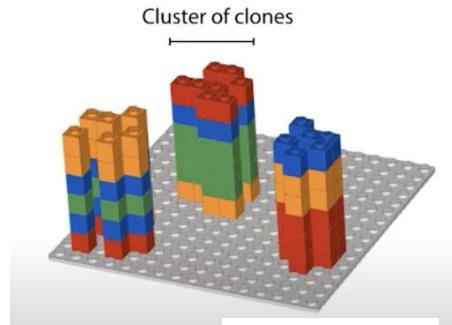
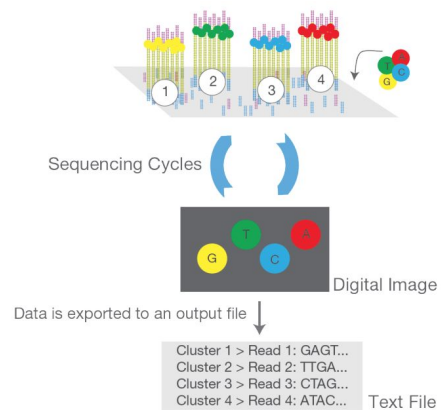
Base calling may not be accurate

Possible causes

- Blocking of synthesis after one nucleotide addition may be inefficient.
- Clusters might not be monoclonal.
- A tile may be out of focus.
- Oil, reagent, etc. on flow cell or imaging component, etc.



=> Need to record quality of each base call.



FastQC: Quality check of sequencing data

- Summarizes quality of base calls
- Any sequences more frequently observed than expected?
- Any sequence biases?
- Any GC biases?
- Checks for presence of known adapters

Examples of FastQC reports

Good Illumina data:

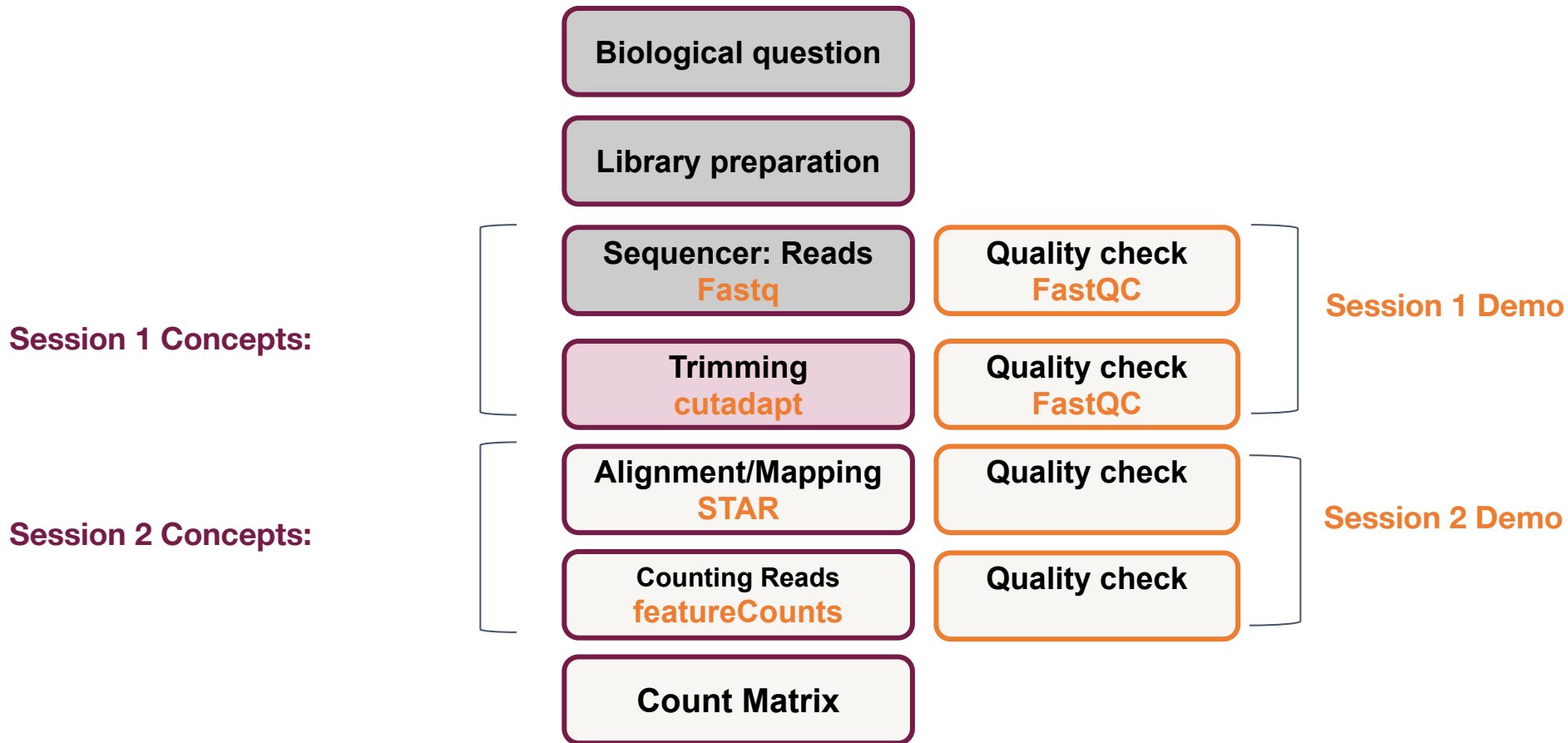
https://www.bioinformatics.babraham.ac.uk/projects/fastqc/good_sequence_short_fastqc.html

Bad Illumina data:

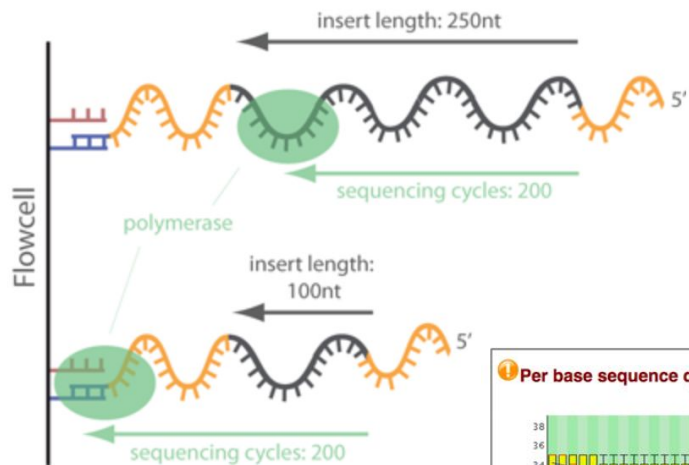
https://www.bioinformatics.babraham.ac.uk/projects/fastqc/bad_sequence_fastqc.html

What if QC gives warn/fail flag?

- Non-normal GC content per read?
 - Normal expected for whole-genome shotgun sequencing.
 - RNA-seq might give different distributions.
- Non-uniform sequence content per nucleotide?
 - First 10-15 nt in RNA-seq often non-uniform.
- High duplication levels or overrepresented sequences?
 - Are they contaminants, e.g. adapters or PCR duplicates?
 - If so, clean up contaminants.
 - Could be attributed to highly abundant transcripts.

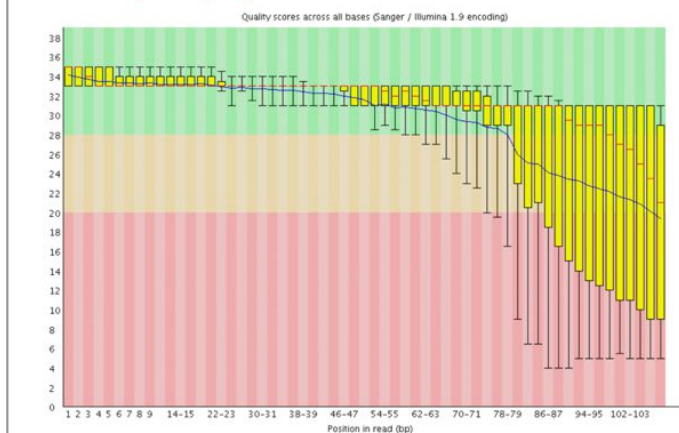


Trimming - Removing adapters



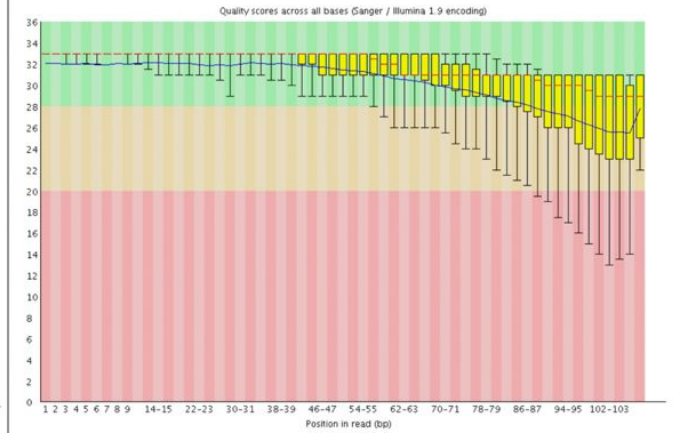
Before trimming

Per base sequence quality



After trimming

Per base sequence quality

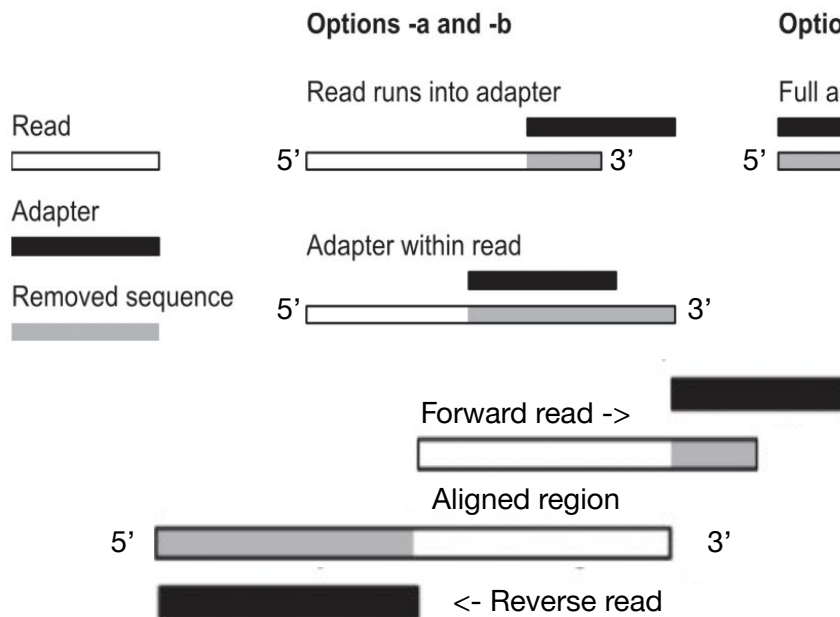


Tools: cutadapt, Trimmomatic

cutadapt removes adapters

Full adapter sequence anywhere	acgtacgtADAPTERacgt
Partial adapter sequence at 3' end	acgtacgtacgtADAP
Full adapter sequence at 3' end	acgtacgtacgtADAPTER

- Search for adapter sequence in read.
- Allow for mismatches in sequence.
- If significant alignment, cut.



Set max error rate (default 1%)
Min overlap (default 3)

The actual error rate is computed as the *number of errors in the match* divided by the *length of the matching part of the adapter*.

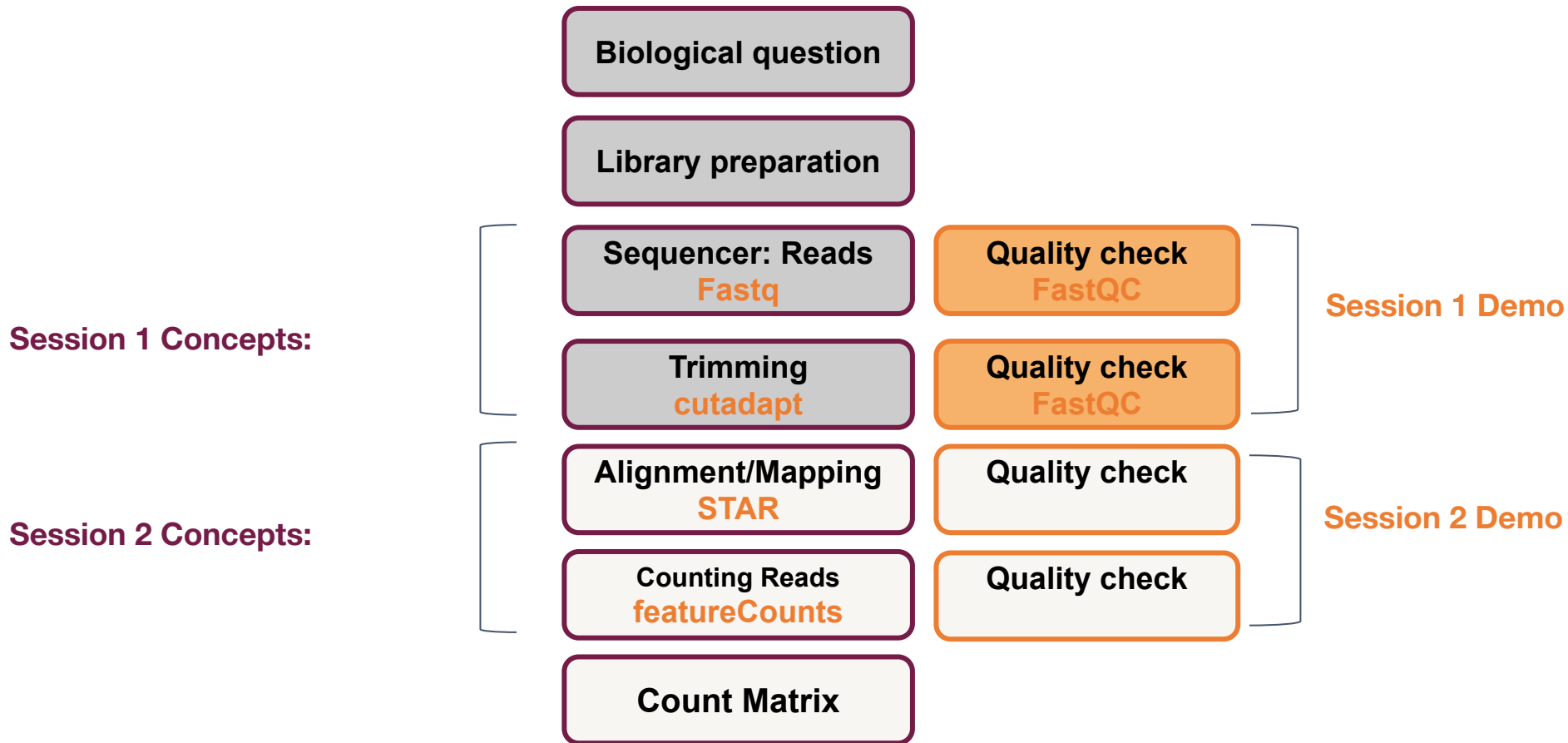
Poll 3

How are you feeling about the content covered so far?

1. Great - I am getting it
2. Okay - I am hanging in there
3. I am learning a lot
4. I need help

**Break
(5 min)**

Please drop a question in the chat or speak up



Bioinformatics software ecosystem

- Tools that “do one thing, and do it well”.
- Tools for this workshop: fastqc, cutadapt, STAR, featureCounts
 - Available via docker hub
 - Some are pre-installed on Wynton; others we can install ourselves
 - Download a *container* with all tools installed; use anywhere
 - Everything can be installed on a laptop

Different platforms for computing

- Web-based platforms
- Graphical User Interface
- Command Line Interface
- etc..

Galaxy: Open source, web-based platform that integrates many tools.

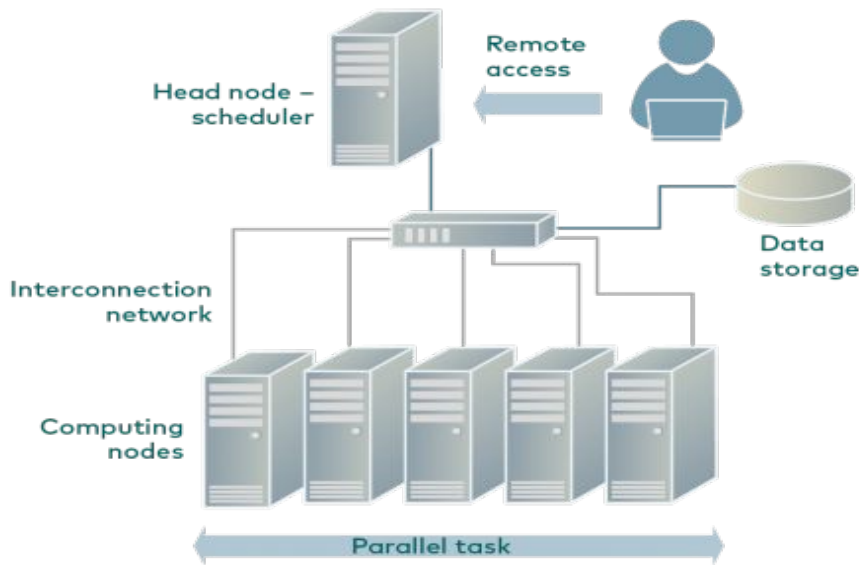
- Free, public, internet accessible resource.
 - <https://usegalaxy.org/>
- Data transfer and data storage are not encrypted.
 - DO NOT UPLOAD PROTECTED DATA!!!

Command line interfaces allow scripting

- **Graphical User Interface**
 - Consists of windows, icons, menus, pointers
 - Not always available for bioinformatics
- **Command Line Interface**
 - Text based
 - Allow automation by scripting
 - Examples
 - Wynton CLI
 - MacOS: Terminal
 - Windows: Command Prompt, PuTTY

Wynton is a high-performance computing (HPC) system for UCSF affiliates

- How to access Wynton?
Visit:
<https://wynton.ucsf.edu/hpc/get-started/access-cluster.html>
- Most universities/ institutions doing data-intensive science have HPC cluster on campus.



(see Description for image source)

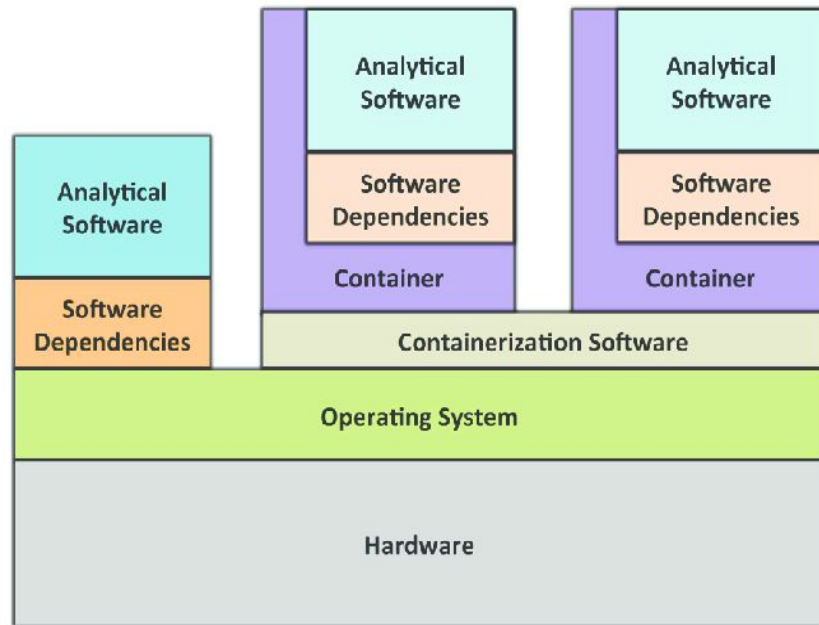
Containers enable reproducibility

- Tools are constantly under development
 - => Many versions around
- Dependencies complicate installations
 - Dependencies are also constantly under development
 - => Many versions around
- Different labs use different programming languages



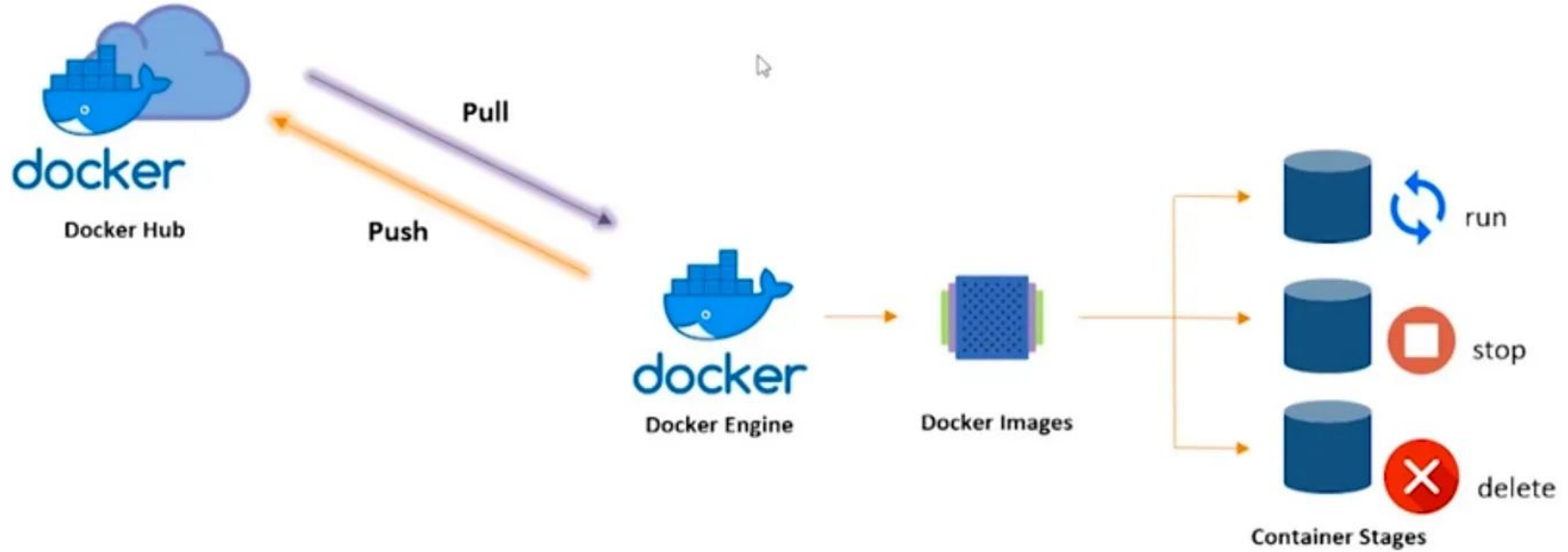
A container is like a computer within a computer

- Containerization software examples:
 - Singularity
 - Docker (has security issues in the context of HPC)
- Containers can be deployed on commercial cloud computing platforms, e.g., Amazon Web Services



(see Description for image source)

Docker Container Lifecycle



Typical command line syntax of bioinformatics tools

```
$ Toolname [options] input
```

```
$ Toolname --paramA 10 -b file.txt -c xyz.fq -outFile pqrs.tuv
```

- Examples:

```
$ fastqc -t 16 -o ./ Bacteria_GATTACA_L001_R1_001.fastq
```

```
$ fastqc *.fastq
```

```
$ cutadapt -a GATTACA -o ./trimmed.fastq input.fastq
```

File paths := Location of file on computer

- `./` is for current working directory
- `../` is for parent directory of current working directory
- `../../` is for parent directory of parent directory of current working directory
- **Example:** If you are currently in data folder, then
`fastqc ./raw.fastq -o ./results/`

```
/home/  
└─ data/  
    └─ raw.fastq  
    └─ results/  
        └─ out.txt
```

Dataset

- **Source Study:** This dataset comes from a study characterizing the transcriptomic changes in human airway smooth muscle (ASM) cells in response to glucocorticoids, which are standard treatments for asthma. ([Himes *et al.*, 2014, PLOS ONE](#))
- **Experimental Conditions:** Primary human ASM cell lines were treated with 1 μ M Dexamethasone (a potent synthetic glucocorticoid) or a control vehicle for 18 hours.
- **Samples in this Workshop:**
 - N61311 cell line - Untreated (Control)
 - N61311 cell line - Dexamethasone treated
- **Workshop Optimization:** To facilitate local analysis and reduce computational load, the raw reads for these samples have been subsampled to include only those mapping to chromosome 16 (chr16).
- **Significance:** The original study used RNA-Seq to identify 316 differentially expressed genes, notably identifying CRISPLD2 as a novel gene that modulates the anti-inflammatory effects of glucocorticoids in the airway

Poll 4

Which operating system will you use for the hands-on section?

1. Mac - Intel chip
2. Mac - Apple chip
3. Windows
4. Linux

Demo

- **Docker desktop installation**

(<https://www.docker.com/products/docker-desktop/>)

(<https://docs.docker.com/desktop/install/windows-install/>)

- **How to access public datasets?**

- Downloading data from GEO

- <https://bioinformaticsworkbook.org/dataAcquisition/fileTransfer/sra.html#gsc.tab=0>
- <https://erilu.github.io/python-fastq-downloader/>

Poll 5

How is the pace?

1. Too fast - please go slower
2. Just right - keep going at the same pace
3. Too slow - please go faster

Session II

See you today at 1.05p

The background features a dark teal color with several wavy, undulating lines in a lighter teal shade. These lines are composed of a fine grid of small dashes or segments, creating a textured, almost 3D effect. The waves flow from the left side towards the right, with some peaks and valleys. The overall aesthetic is modern and scientific.

GLADSTONE INSTITUTES

Introduction to RNA-seq data analysis

Michela Traglia, Ayushi Agrawal
Bioinformatics Core, GIDB
September 28-29, 2026

GLADSTONE
INSTITUTES

Workshop outline

Session I

- RNA-seq analysis overview
- Understanding the sequencer output
- Quality checks
- Trimming of adapters

Break

- Introduction to docker and setup
- Demo - QC steps

Session II

- Mapping the read to reference genome
- From alignment to counting features

Break

- Demo - from Mapping to the count matrix

Session III

- Filtering low counts genes
- Dimensionality reduction
- Normalization

Break

- Model fitting
- DEG and visualization

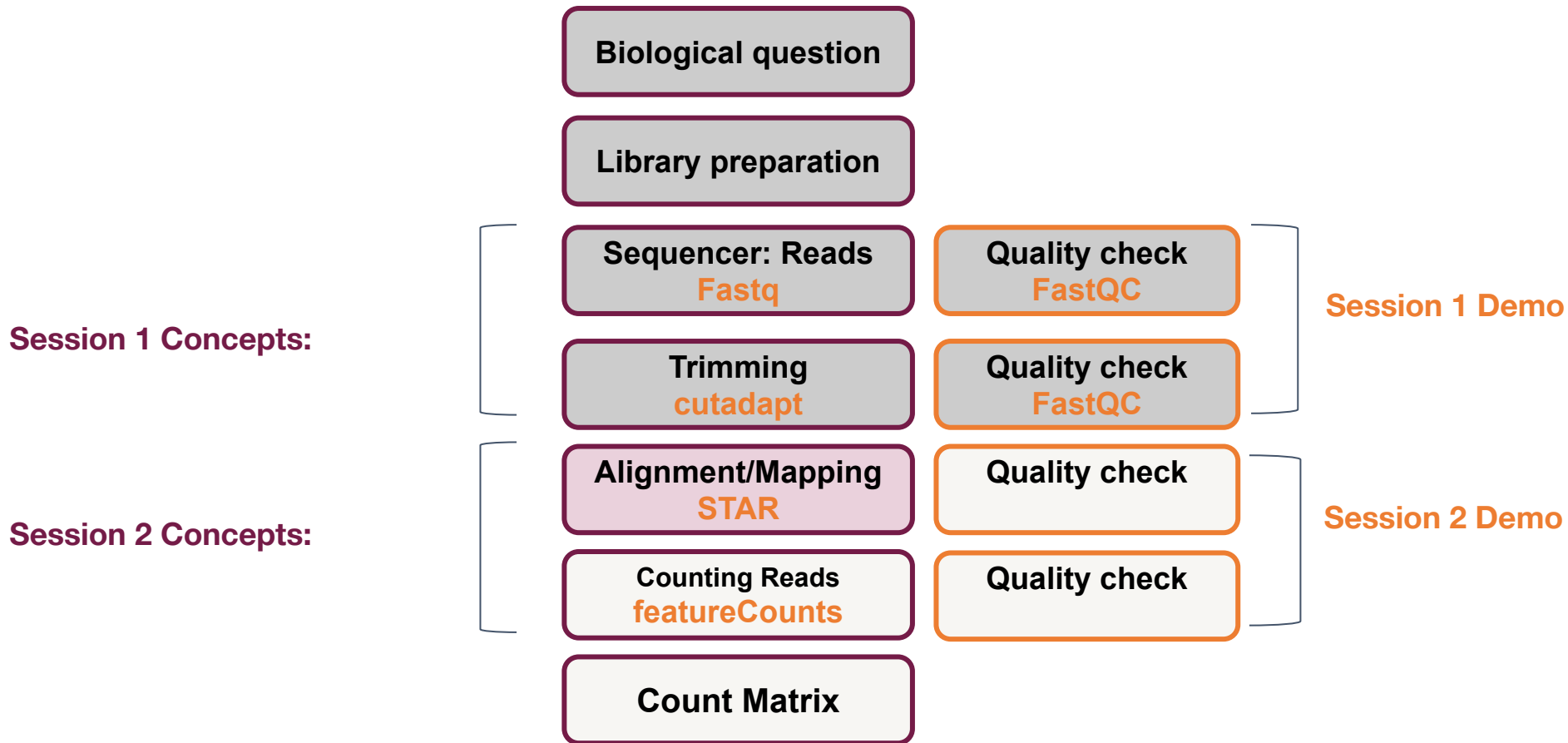
Break

- Demo - from count matrix to DEG

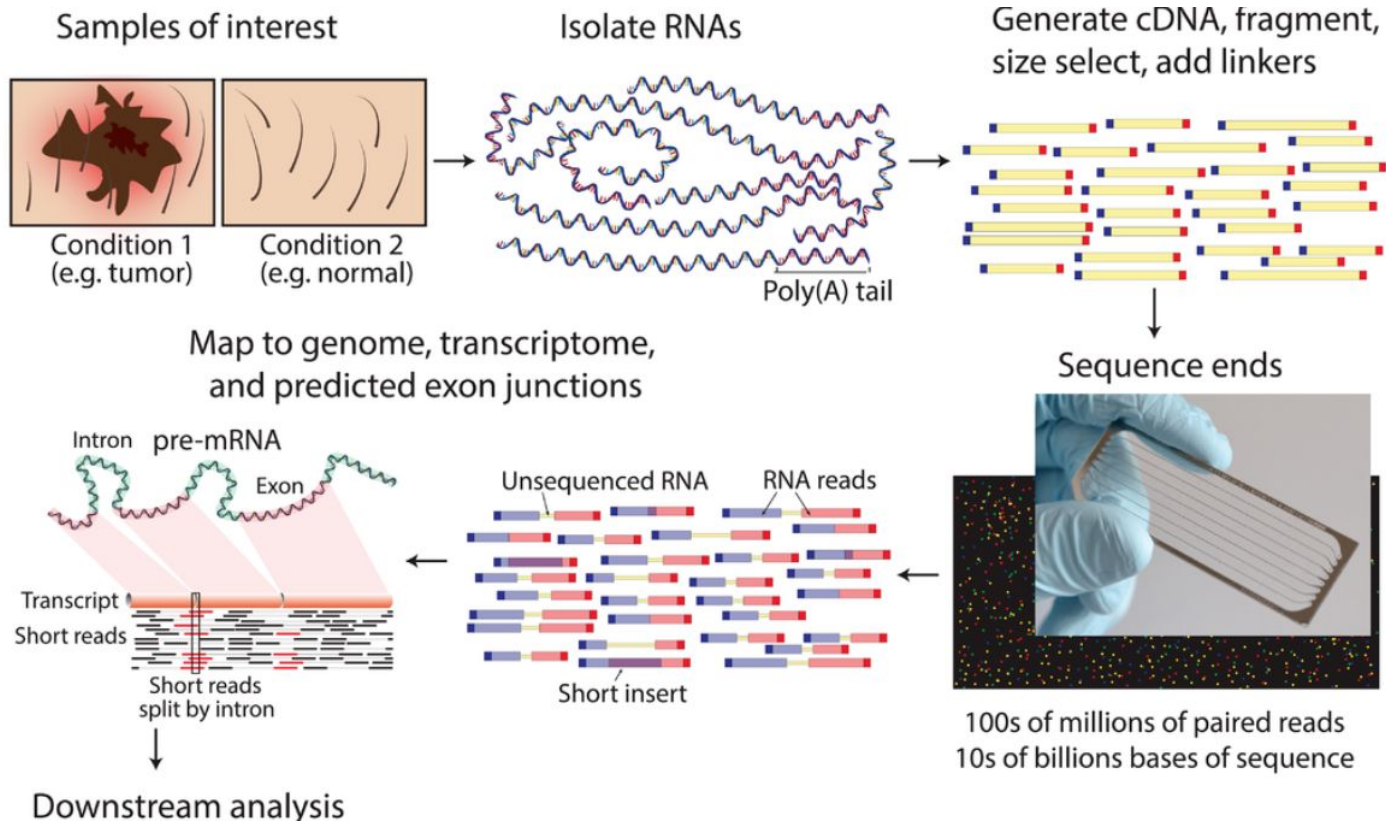
Session 2

Alignment/mapping and feature counts

Tools: Docker, STAR, featureCounts

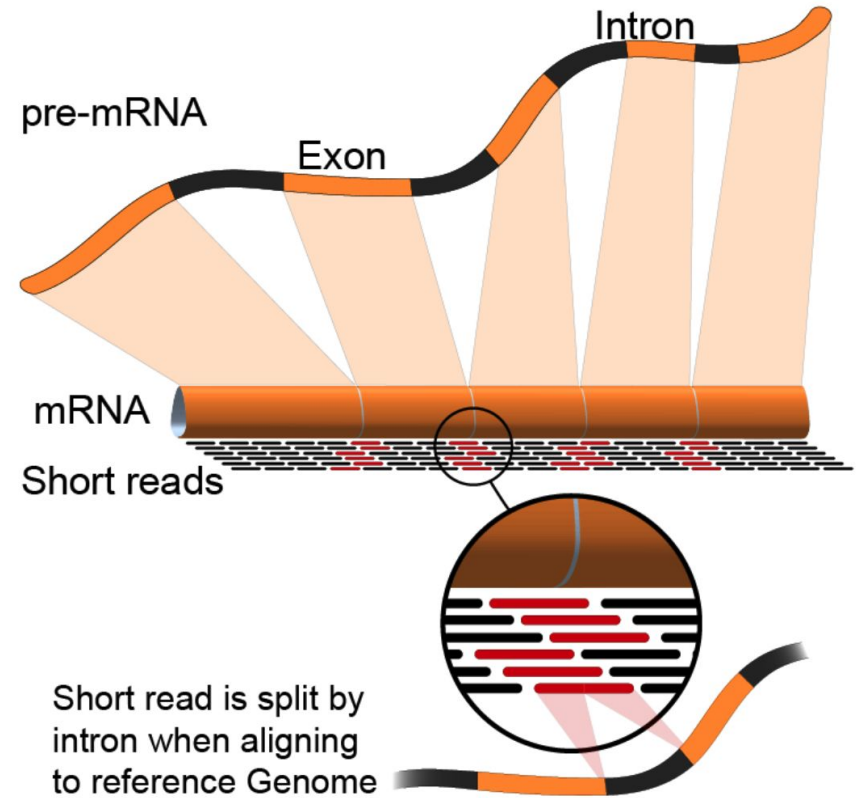


Bulk mRNA-seq - summary



Alignment to genome : challenges

- Reads from junctions: one part of it maps to one exon and the other half maps to the other exon
- Reference sequences can be very long (~3 billion bp for humans).
- Order of 100 million reads to be mapped.
- Need to account for splicing.
- Allow for PCR artifacts/sequencing errors.



Many tools

- STAR, HISAT2, TopHat2, bowtie2, salmon, kallisto, etc.
- Differences in speed and memory requirement
- Some are aligner and some are pseudoaligners
- Pros and cons of each:
 - Example: Some handle spliced alignment, others do not.
 - ...

[Comparison across tools](#)

Main characteristics of alignment tools

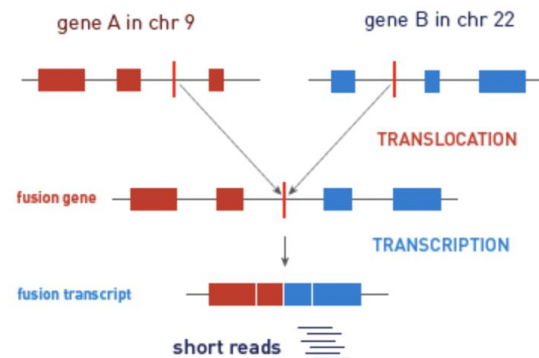
- TopHat2: very slow, no complex splicing, not maintained
- HISAT2: similar to STAR, low memory usage, fast, standard for genome alignment
- **STAR**: very fast and very accurate, high memory, standard for splicing junction
- Bowtie2: no splicing alignment

Alignment/Mapping: STAR tool

STAR (Spliced Transcripts Alignment to a Reference (STAR)) is popular for RNA-Seq data because it

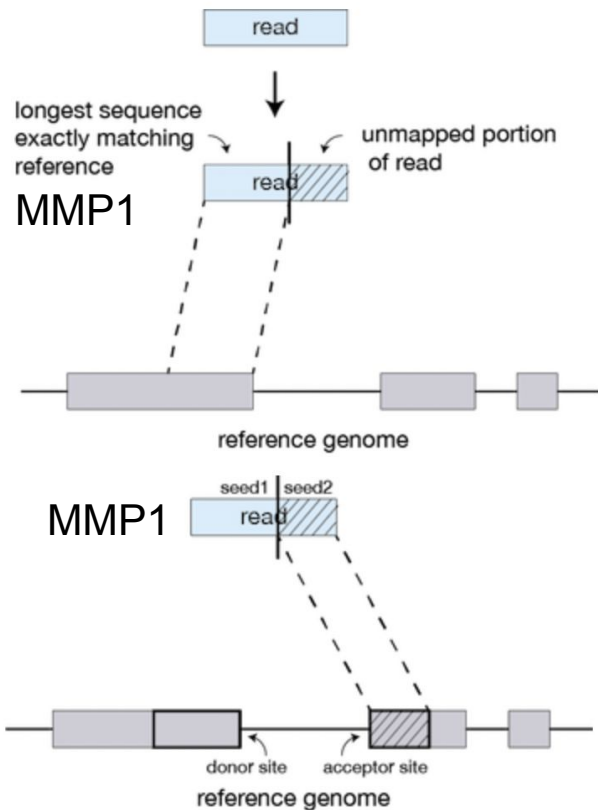
- does unbiased de novo detection of canonical junctions
- can discover non-canonical splices
- can discover chimeric (fusion) transcripts

Tools: STAR, HISAT2, TopHat2, bowtie2, salmon, kallisto

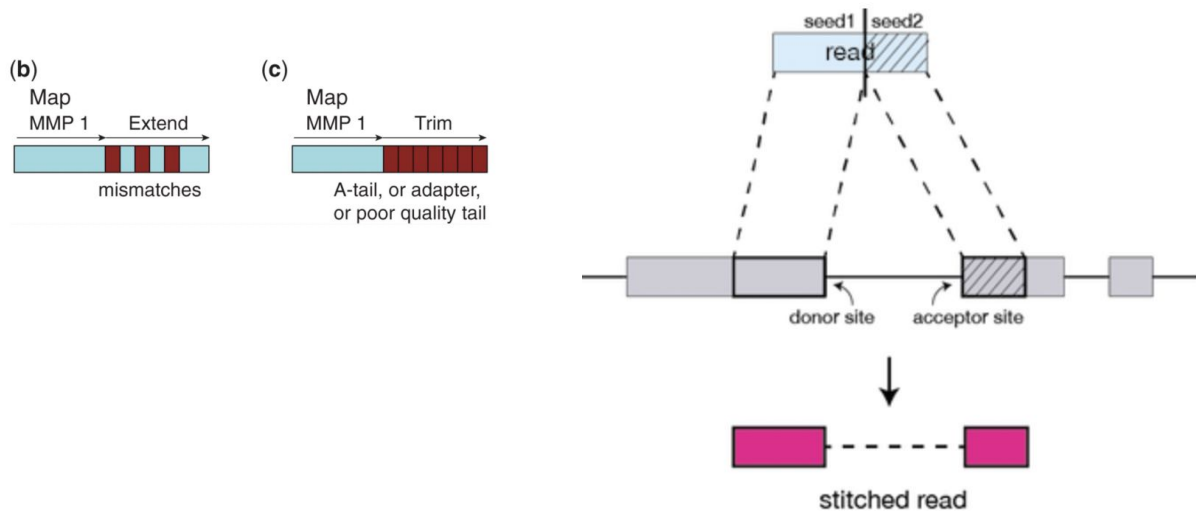


Alignment/Mapping: How does STAR work?

Step 1: Seed search



Step 2: Stitching and scoring



MMP = Maximum Mappable Prefix
Anchor seeds in a genomic region

Alignment/Mapping: Inputs needed

- Reads to align
 - FASTQ file after cleaning (trimming adapters).
- Reference sequence to align to
 - Example – “rDNA_sequence.fasta”
 - FASTA format. Two lines per sequence.
 - Starting with “>”, followed by sequence name/identifier.
 - Sequence.
 - File extensions: .fasta, .fa, .txt.
 - Examples of acceptable genome sequence files for STAR:
 - ENSEMBL: files marked with *.dna.primary.assembly*
 - GENCODE: files marked with *PRI* (primary). Strongly recommended for mouse and human.

From the alignment to SAM format

Alignment

```
Coord      12345678901234 5678901234567890123456789012345
ref        AGCATGTTAGATAA**GATAGCTGTGCTAGTAGGCAGTCAGCGCCAT
```

```
+r001/1      TTAGATAAAGGATA*CTG
+r002        aaaAGATAA*GGATA
+r003        gcctaAGCTAA
+r004                ATAGCT.....TCAGC
-r003                ttagctTAGGC
-r001/2                CAGCGGCAT
```

$$MAPQ = -10 * \log_{10}(P_{map_loc_wrong})$$

```
@HD VN:1.6 SO:coordinate
```

```
@SQ SN:ref LN:45 CIGAR string
```

```
r001  99 ref  7 30 8M2I4M1D3M = 37 39 TTAGATAAAGGATACTG *
r002   0 ref  9 30 3S6M1P1I4M * 0 0 AAAAGATAAGGATA *
r003   0 ref  9 30 5S6M * 0 0 GCCTAAGCTAA * SA:Z:ref,29,-,6H5M,17,0;
r004   0 ref 16 30 6M14N5M * 0 0 ATAGCTTCAGC *
r003 2064 ref 29 17 6H5M * 0 0 TAGGC * SA:Z:ref,9,+,5S6M,30,1;
r001 147 ref 37 30 9M = 7 -39 CAGCGGCAT * NM:i:1
```

SAM

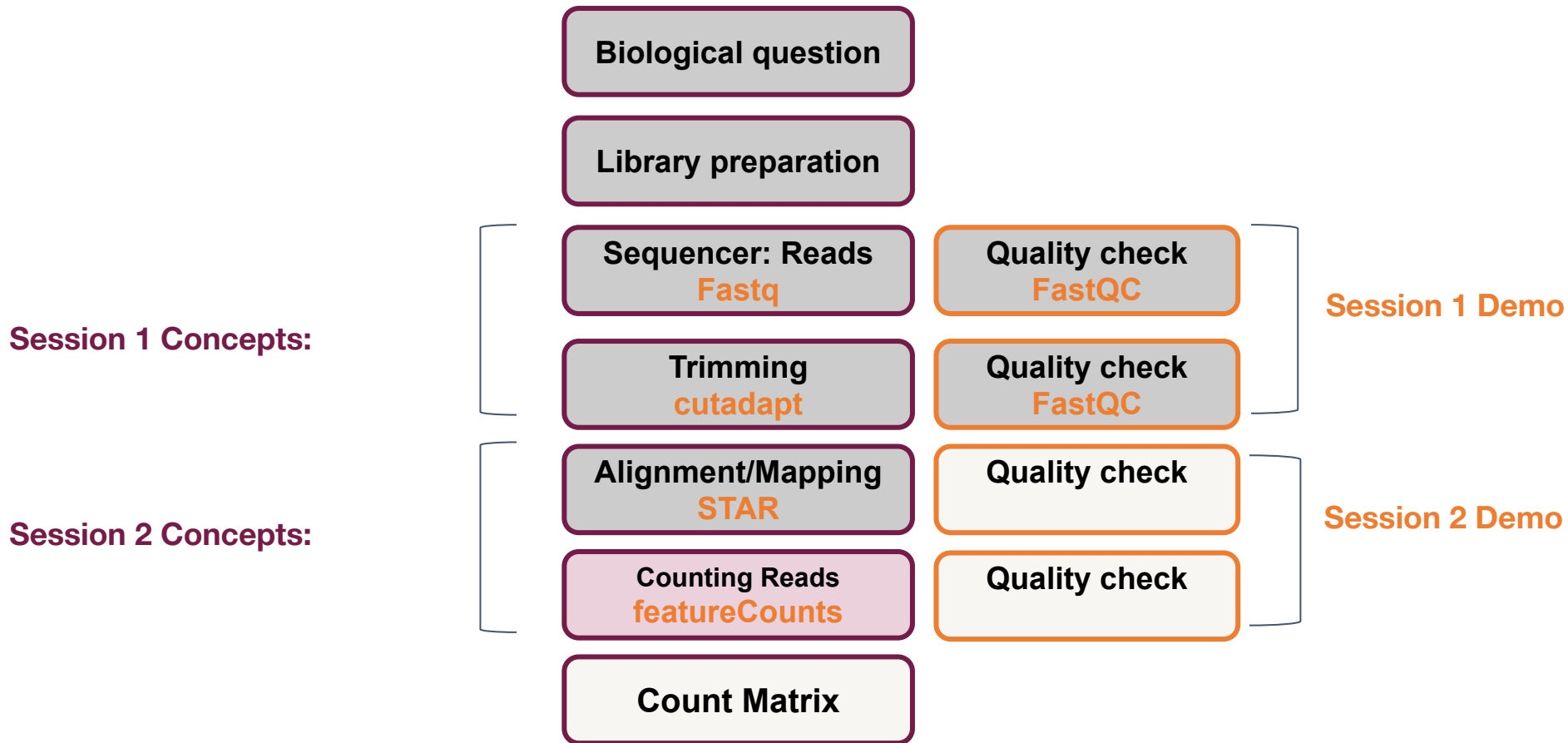
Sequence Alignment/Map (SAM) format

Alignments in SAM format - a summary of mapping statistics

- How many reads mapped?
- How many unmapped?
- ...
- First few lines contain metadata about alignments.
 - These lines start with “@”.
 - Example – version of file format, sorting order of alignments, grouping, etc.
- After header, a table of alignments of each read to the genome.
- Alignment reports often very large files.
- Binary Alignment/Map (BAM) extension used for compressed SAM files.
- Indexed BAM -> BAI

Pseudo-aligners identify the transcript that originated the read

- Much, much faster
- Less memory intensive than STAR
- They are called "pseudo" aligners because they do not perform the full alignment of each sequencing read to a reference genome or transcriptome.
- Pseudoaligners use a reference transcriptome to create a set of potential transcript sequences or exonic regions, respectively. Then, they map the reads to these sequences without attempting to align each read to its exact position in the reference.
- Examples of tools: **Salmon, Kallisto**



featureCounts

- Input:
 - Alignment BAM file
 - Annotation SAF/GFF/GTF file

GeneID	Chr	Start	End	Strand
497097	chr1	3204563	3207049	-
497097	chr1	3411783	3411982	-
497097	chr1	3660633	3661579	-
100503874	chr1	3637390	3640590	-

aligned read:
start: 113217600 end: 113217650

SAF



GTF

chr1	unknown	exon	113217048	113217252	.	+	.	gene_id	"MOV10";p_id	"P5535";transcript_id	"NM_001130079"
chr1	unknown	exon	113217048	113217351	.	+	.	gene_id	"MOV10";p_id	"P5535";transcript_id	"NM_020963"
chr1	unknown	exon	113217470	113217671	.	+	.	gene_id	"MOV10";p_id	"P5535";transcript_id	"NM_001130079"
chr1	unknown	CDS	113217535	113217671	.	+	0	gene_id	"MOV10";p_id	"P5535";transcript_id	"NM_001130079"
chr1	unknown	start_codon	113217535	113217537	.	+	.	gene_id	"MOV10";p_id	"P5535";transcript_id	"NM_001130079"

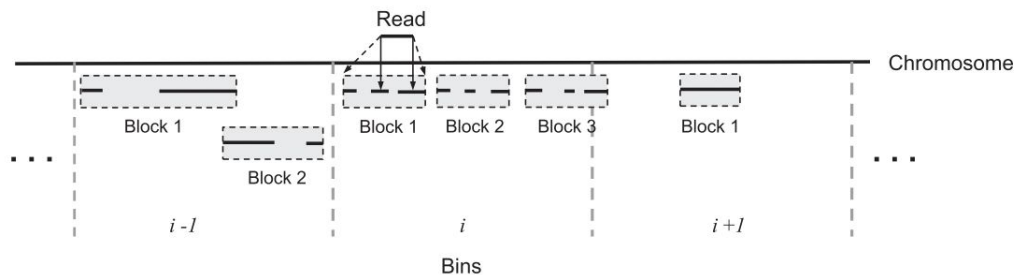
↑
feature type

↑
feature

meta-feature is the aggregation of a set of features with the same gene_id

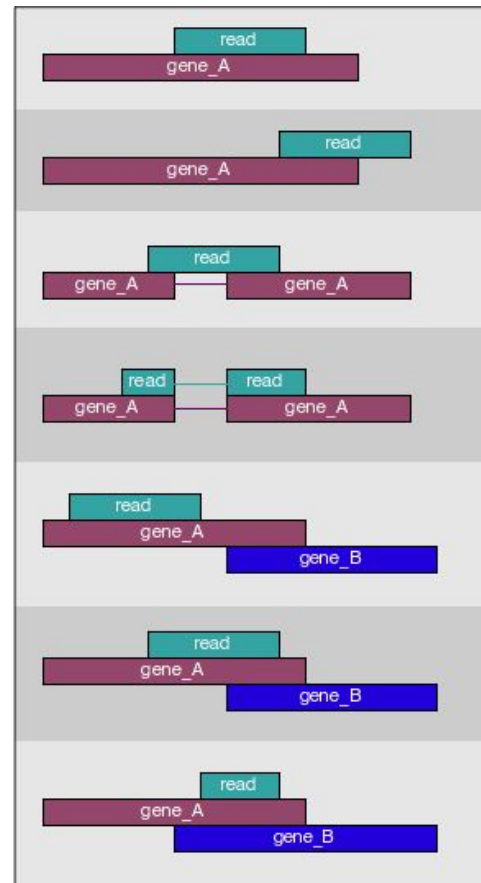
How many reads overlap annotated regions?

Use [featureCounts](#)



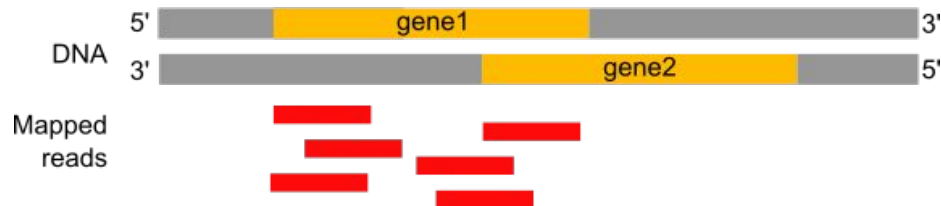
Narrow down the genomic region that could contain features overlapping with the query read

Hierarchical data structure - features within blocks within bins

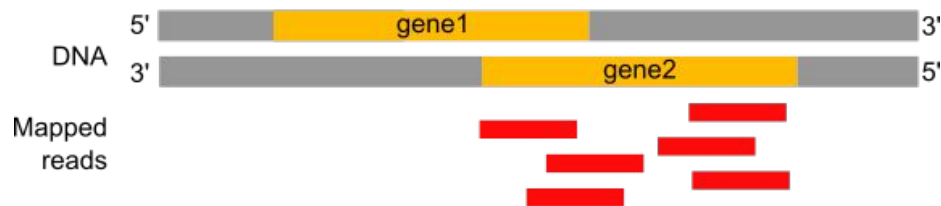


Estimation of the strandness - directionality of the RNA molecule

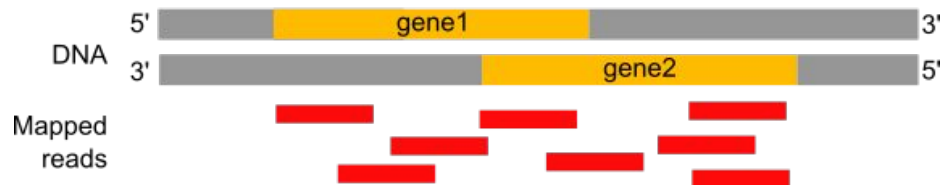
Stranded library: forward



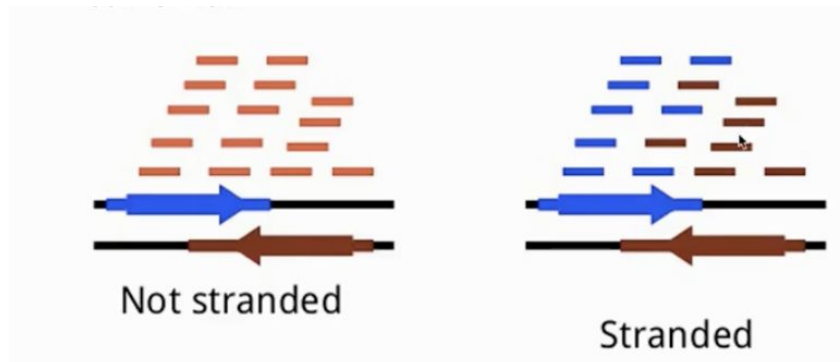
Stranded library: reverse



Unstranded library



- In a stranded forward library, reads map mostly on the genes located on forward strand (here gene1).
- With stranded reverse library, reads map mostly on genes on the reverse strand (here gene2).
- With unstranded library, reads map on genes on both strands.



featureCounts output

- count matrix (text file)
- a summary file

Each column is a sample

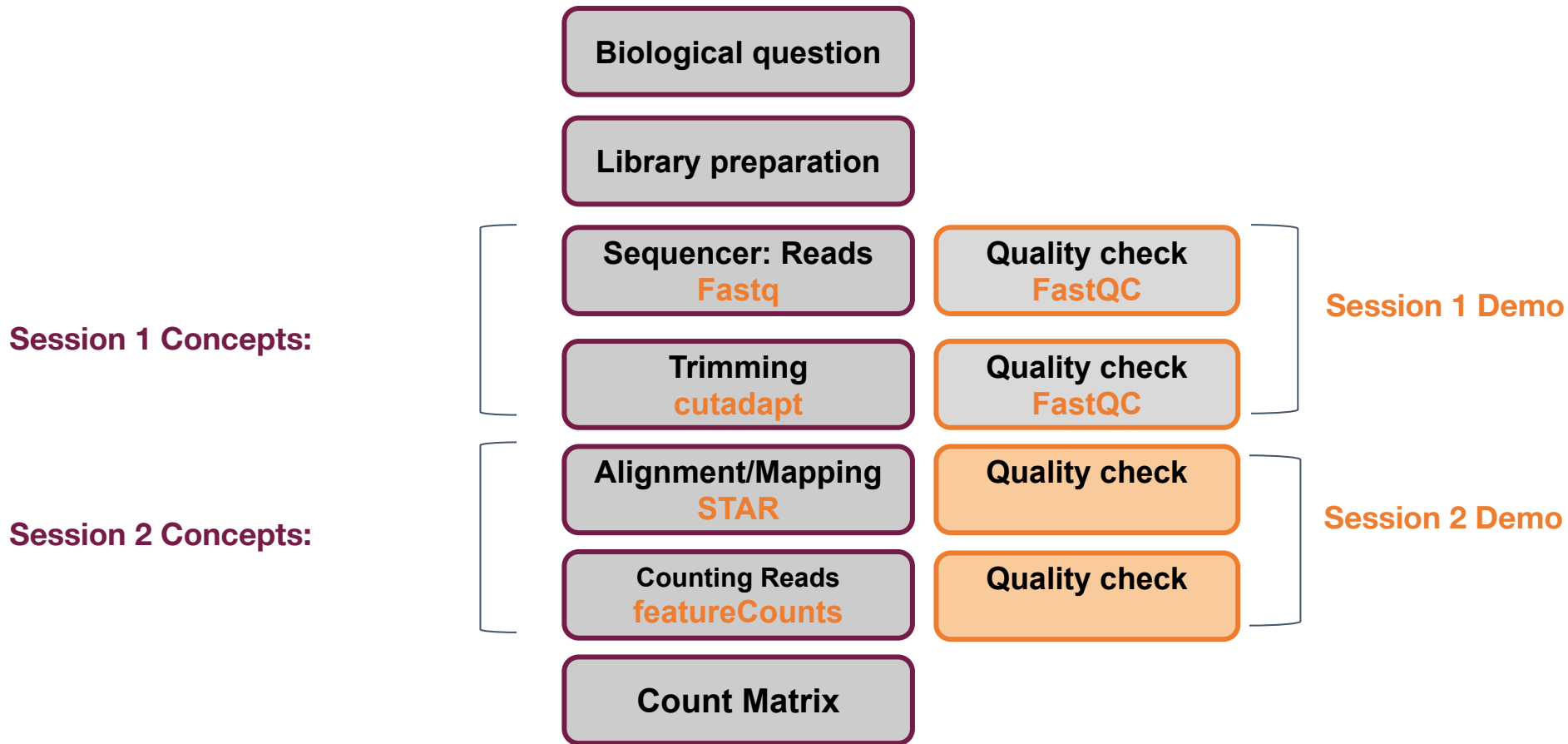
Each row is a gene

GENE ID	KD.2	KD.3	OE.1	OE.2	OE.3	IR.1	IR.2	IR.3
1/2-SBSRNA4	57	41	64	55	38	45	31	39
A1BG	71	40	100	81	41	77	58	40
A1BG-AS1	256	177	220	189	107	213	172	126
A1CF	0	1	1	0	0	0	0	0
A2LD1	146	81	138	125	52	91	80	50
A2M	10	9	2	5	2	9	8	4
A2ML1	3	2	6	5	2	2	1	0
A2MP1	0	0	2	1	3	0	2	1
A4GALT	56	37	107	118	65	49	52	37
A4GNT	0	0	0	0	1	0	0	0
AA06	0	0	0	0	0	0	0	0
AAA1	0	0	1	0	0	0	0	0
AAAS	2288	1363	1753	1727	835	1672	1389	1121
AACS	1586	923	951	967	484	938	771	635
AACSP1	1	1	3	0	1	1	1	3
AADAC	0	0	0	0	0	0	0	0
AADACL2	0	0	0	0	0	0	0	0
AADACL3	0	0	0	0	0	0	0	0
AADACL4	0	0	1	1	0	0	0	0
AADAT	856	539	593	576	359	567	521	416
AAGAB	4648	2550	2648	2356	1481	3265	2790	2118
AAK1	2310	1384	1869	1602	980	1675	1614	1108
AAMP	5198	3081	3179	3137	1721	4061	3304	2623
AANAT	7	7	12	12	4	6	2	7
AARS	5570	3323	4782	4580	2473	3953	3339	2666
AARSA	4454	2727	3381	3121	1540	3480	2874	1657

Break (5 min)

Need help? Please drop a question in the chat or speak up

RNA-seq - analysis workflow



Demo using Docker

- Run STAR to align the reads to the reference genome
- Run featureCounts to tally counts

Real data is complex

- Session I and II provides an introduction to typical RNA-seq analysis steps using an artificial dataset.

Real data might need additional analyses choices.

- Possible challenges with real data: (What we did not cover today)
 - What to do you if only 50% of reads align to reference?
 - What to do if FastQC reports unusual GC content?
 - What to do if the reference genome is incomplete?
 - How to deal with more complicated experimental designs?
 - How many replicates to use for RNA-seq?
- Some analysis choices need experience. Consult with the [Gladstone Bioinformatics core](#) for such scenarios and data.

Production-ready pipelines: nf-core/rnaseq

What you will learn today:

Individual tools (FastQC, cutadapt, STAR, featureCounts)

What we use at Gladstone Bioinformatics Core:

Automated pipelines (nf-core/rnaseq)

- Runs all steps automatically
- Handles batch processing (100+ samples)
- Built-in best practices
- Comprehensive quality reports

 nf-core/rnaseq documentation:
<https://nf-co.re/rnaseq>

Session I and II

Take-home messages

- 1) Each steps of the analysis can be affected by some kind of bias - Check the quality after each step!
- 2) Know the tools and how they work. Use best practices for the analysis.

What we covered so far:

- 1) Common tools, e.g., fastqc, cutadapt, STAR
- 2) Common file formats, e.g., FASTQ, FASTA, SAM, GTF, BAM
- 3) Analysis on your computer using docker

**If you are not able to attend
Session III tomorrow at 1.05p**

Please take the survey

<https://www.surveymonkey.com/r/bioinfo-training26>

Introduction to RNA-seq data analysis

Michela Traglia, Ayushi Agrawal
Bioinformatics Core, GIDB
September 28-29, 2026

GLADSTONE
INSTITUTES

Working material for this workshop

Session I-II-III

1. Slides
2. Session I-II - Hands-on materials
3. **Session III** - Hands-on materials

Hands-on session III files

- R
- **2026Fall_RNAseq_SessionIII_handson.R** #each step in
 - targets.txt #sample sheet
 - DE_result_handson.txt #results table
 - GSE60450_Lactation-GenewiseCounts.txt.gz #count matrix

Session 3

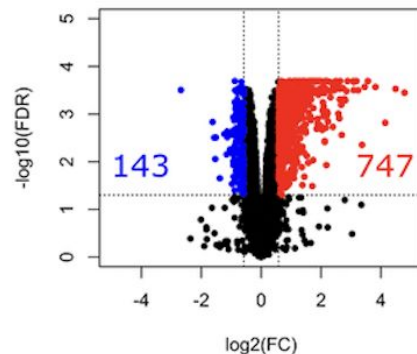
From count matrix to DEGs

From the count matrix to the DEG

Each column is a sample

Each row is a gene

GENE ID	KD.2	KD.3	OE.1	OE.2	OE.3	IR.1	IR.2	IR.3
1/2-SBSRNA4	57	41	64	55	38	45	31	39
A1BG	71	40	100	81	41	77	58	40
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A1CF	0	1	1	0	0	0	0	0
A2LD1	146	81	138	125	52	91	80	50
A2M	10	9	2	5	2	9	8	4
A2ML1	3	2	6	5	2	2	1	0
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A4GNT	0	0	0	0	1	0	0	0
AA06	0	0	0	0	0	0	0	0
AAA1	0	0	1	0	0	0	0	0
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AADACL4	0	0	1	1	0	0	0	0
AADAT	856	539	593	576	359	567	521	416
AAGAB	4648	2550	2648	2356	1481	3265	2790	2118
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AANAT	7	7	12	12	4	6	2	7
AARS	5570	3323	4782	4580	2473	3953	3339	2666
AARSD	4454	2323	3281	3121	1340	2488	2074	1657



Workshop outline

Session I

- RNA-seq analysis overview
- Understanding the sequencer output
- Quality checks
- Trimming of adapters

Break

- Introduction to docker and setup
- Demo - QC steps

Session II

- Mapping the read to reference genome
- From alignment to counting features

Break

- Demo - from Mapping to the count matrix

Session III

- Filtering low counts genes
- Dimensionality reduction
- Normalization

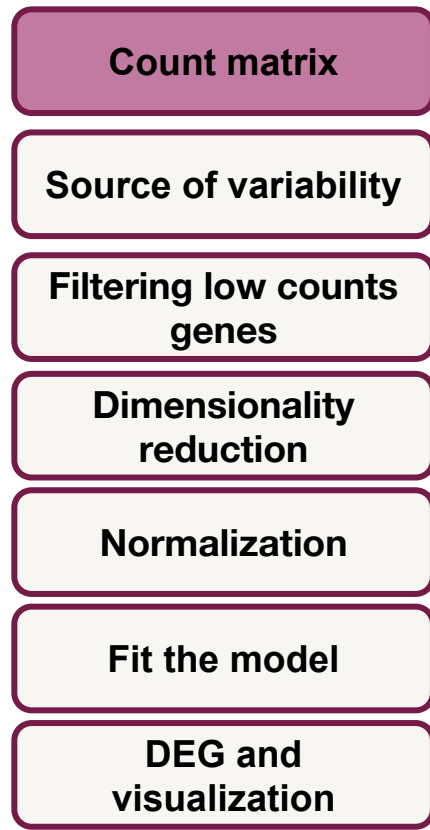
Break

- Model fitting
- DEG and visualization

Break

- Demo - from count matrix to DEG

Session 3 Concepts:



Session 3 Demo

Reference for the workshop



SOFTWARE TOOL ARTICLE

REVISED From reads to genes to pathways: differential expression analysis of RNA-Seq experiments using Rsubread and the edgeR quasi-likelihood pipeline [version 2; referees: 5 approved]

Yunshun Chen^{1,2}, Aaron T. L. Lun ³, Gordon K. Smyth ^{1,4}

¹The Walter and Eliza Hall Institute of Medical Research, Parkville, Victoria, 3052, Australia

²Department of Medical Biology, The University of Melbourne, Victoria, 3010, Australia

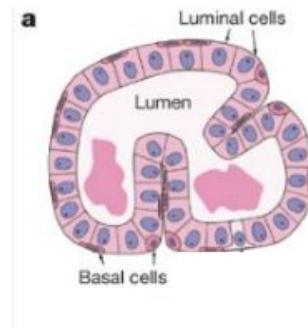
³Cancer Research UK Cambridge Institute, University of Cambridge, Li Ka Shing Centre, Cambridge, UK

⁴Department of Mathematics and Statistics, The University of Melbourne, Victoria, 3010, Australia

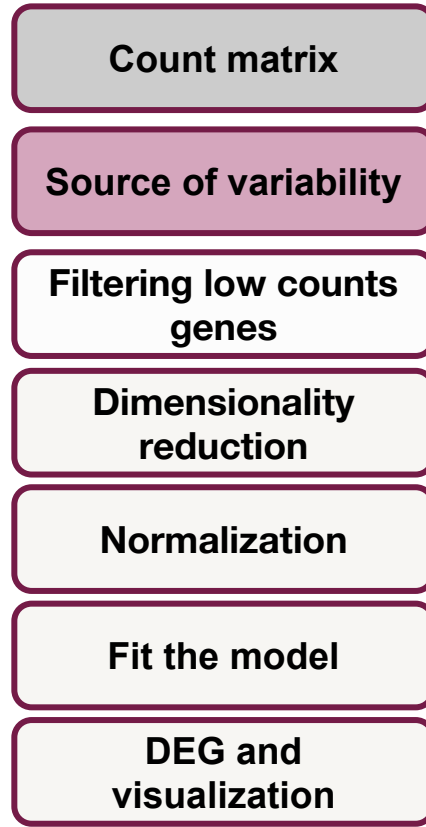
Dataset

Transcriptome analysis of luminal and basal cell subpopulations in the lactating versus pregnant mammary gland

- GEO (gene expression omnibus) accession: **GSE60450**
<https://www.ncbi.nlm.nih.gov/gds>
- Tissue of origin: Mammary glands of mouse
- Cell types: Basal stem-cell enriched cells (B) and committed luminal cells (L)
- Biological conditions: Virgin, Lactating (2 day) and Pregnant (18.5 day)
- # of groups: 2 cell types (B/L) x 3 conditions (V/L/P) = 6 groups
- # of replicates: 2 of each group
- Illumina Hiseq sequencer - about 30 million 100bp single-end reads for each sample.

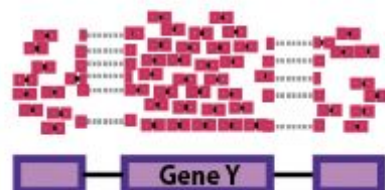
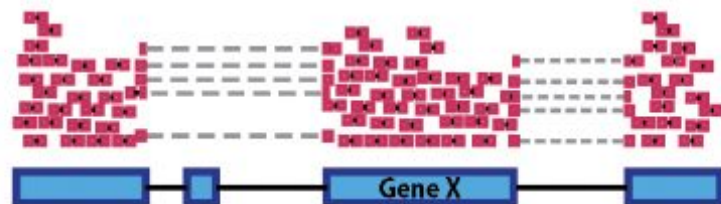


Session 3 Concepts:

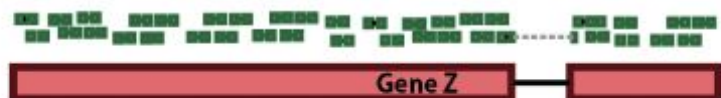
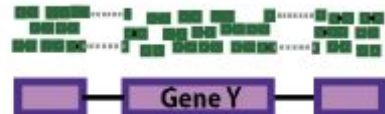
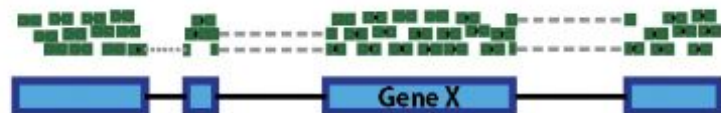


Session 3 Demo

Sample A Reads

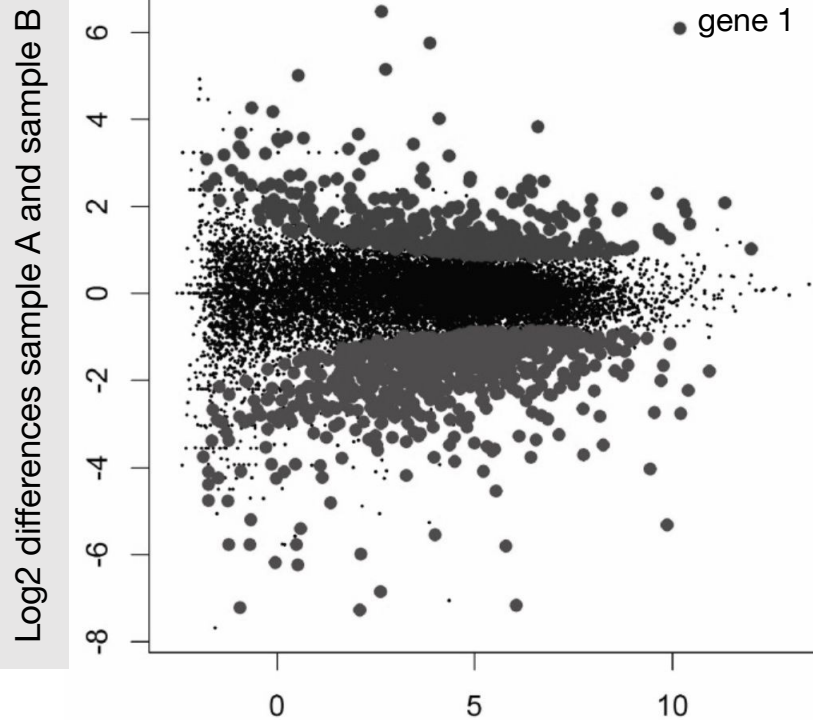


Sample B Reads



Source of variability

Visualize replicates using s MD or MA plot

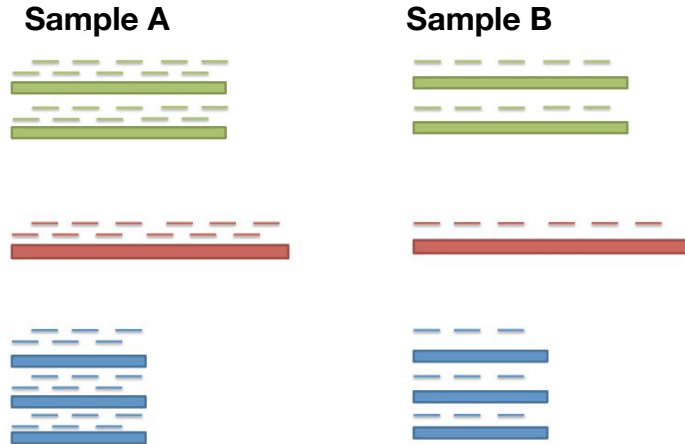


Sample A - expression gene 1
Sample B - expression gene 1

Average expression between sample A and sample B

We need to detect and correct for the sources of bias

Sample A has more counts assigned to these 3 genes compared to sample B



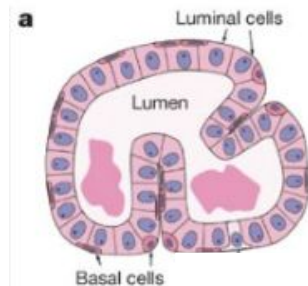
Source of bias: Sequencing depth (or library size or total number of reads) of each sample

Different sequence depths may cause variation in counts

Variation in sequencing depths => Need to normalize counts

Group	Total counts
B.virgin	23085177
B.virgin	21628857
B.pregnant	23919152
B.pregnant	22490570
B.lactating	21382233
B.lactating	19884434

Group	Total counts
L.virgin	20213223
L.virgin	21509988
L.pregnant	22073815
L.pregnant	21837341
L.lactating	24638939
L.lactating	24581591



Library size about 20M

Sequence depth/Library size - Count Per Million mapped reads

$$\text{RPM or CPM} = \frac{\text{Number of reads mapped to gene} \times 10^6}{\text{Total number of mapped reads}}$$

Sequenced one library with 5 million(M) reads.

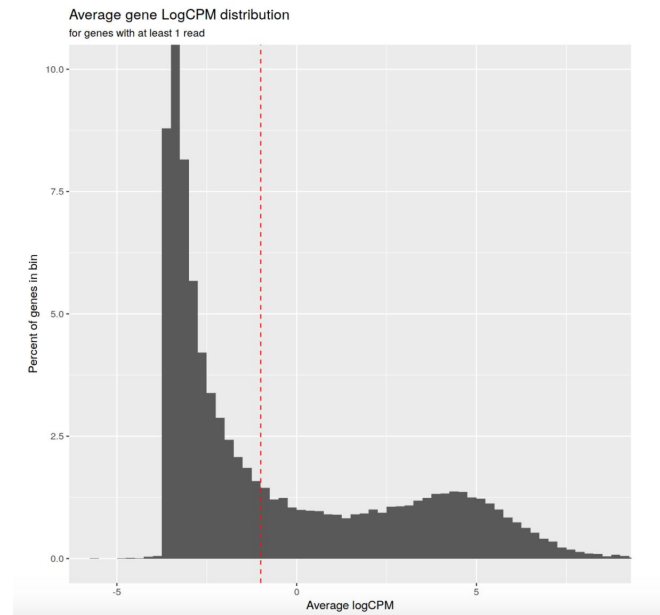
Total 4 M matched to the genome sequence

5000 reads matched to a given gene

$$\text{RPM or CPM} = \frac{5000 \times 10^6}{4 \times 10^6} = 1250$$

Filter on count-per-million (CPM) values to avoid favoring genes that are expressed in larger libraries over those expressed in smaller libraries

A gene at least 10–15 counts in at least some libraries before it is considered to be expressed -> Identifying the CPM that corresponds to 10-15 counts

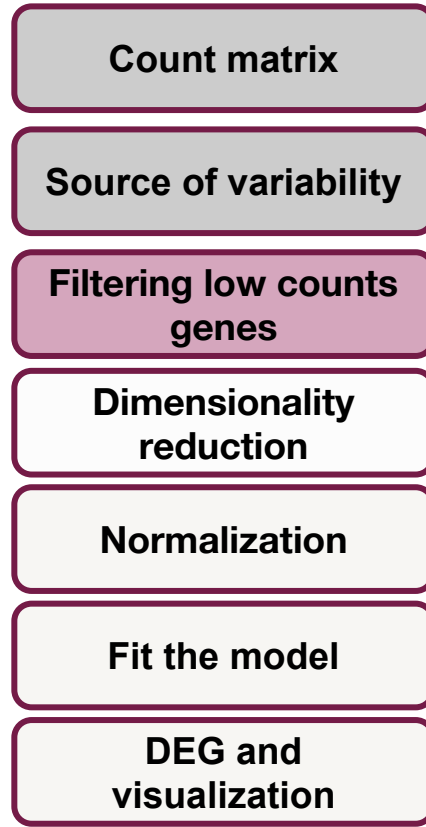


logCPM values allow

- Accounting for sequencing depth
- Filtering out low counts genes
- Visualizing inter sample variability

Not the correct normalization for DEG analysis

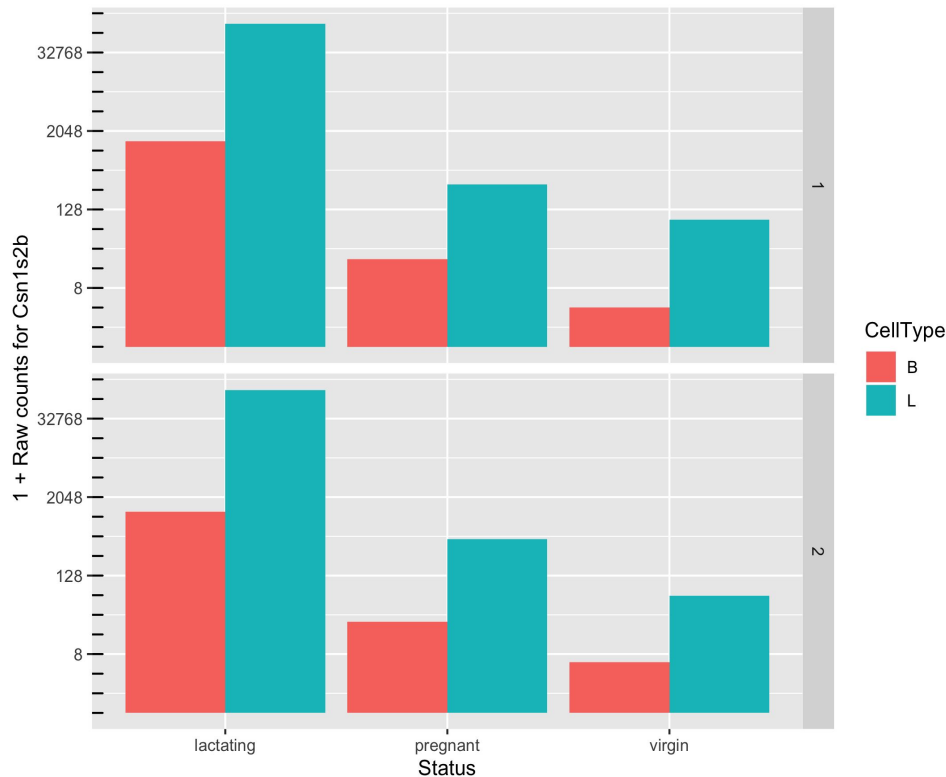
Session 3 Concepts:



Session 3 Demo

Filtering

Potential biological questions



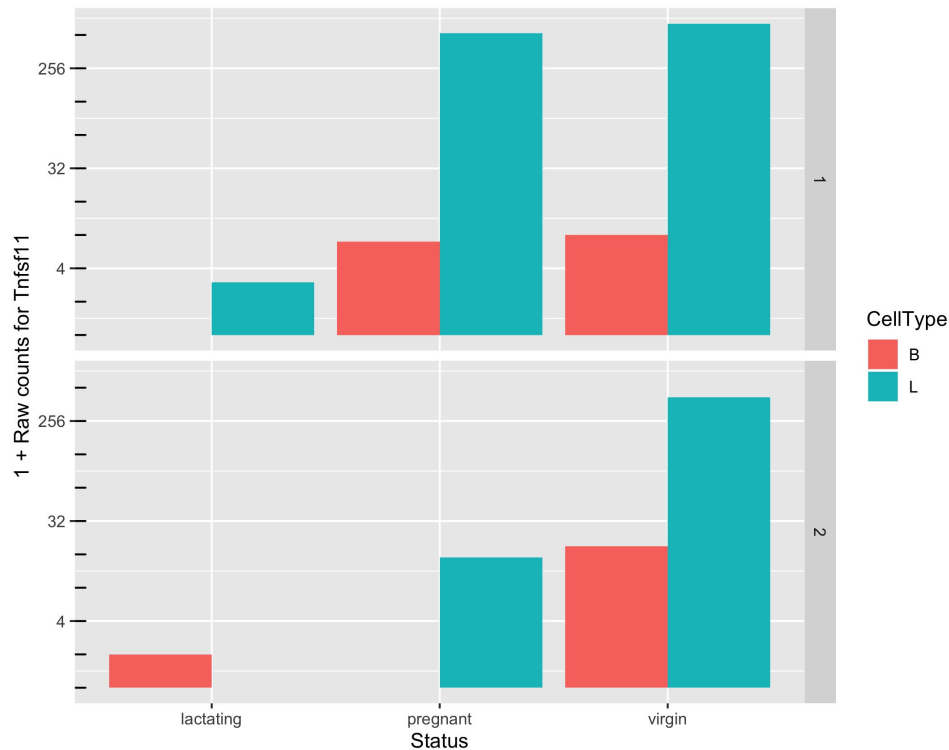
Which comparisons are we interested in?

Example:

1. B vs L,
2. B.lactating vs L.pregnant,
3. ...
4. All of them

Or not interested in comparisons but in gene expression of one sample

Issue: genes with low counts



Can we make reliable inferences for genes with very low counts? What should we consider “very low”?

“Uninteresting” genes

Filtering

- *Biological point of view*: minimal expression level of a gene -> translation into a protein -> biologically relevant
- *Statistical point of view*: low counts -> not enough statistical evidence.

Genes with consistently low counts are very unlikely be assessed as significantly DE

Genes with extreme count outlier

Genes with zero counts

Genes with low mean normalized counts

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG00000000003	67	44	87	40	1138
ENSG00000000005	0	0	0	0	0
ENSG000000000419	467	515	621	365	587
ENSG000000000457	260	211	263	164	245
ENSG000000000460	2	5	1	0	1

Using cpm to filter out genes

Need to correct for multiple testing

P value represents the chance that we may be wrong in calling something significantly differential. Example:

- $P = 0.01$ means 1% chance that we may be wrong.

More than 20k genes under consideration

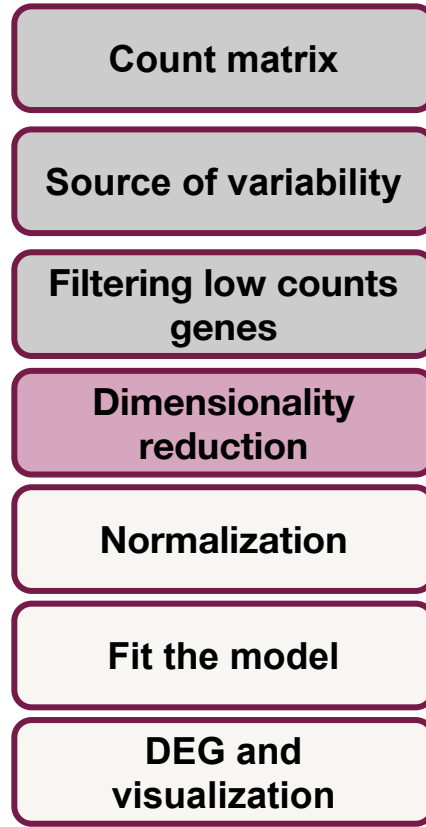
=> if a certain difference in expression levels has only 1% chance of happening given the null model, it might be observed for 200 genes even if the null model were true for all the genes.

=> 200+ false positives

Hence, there is a need to adjust the p-values.

- The more genes we test, the more we must adjust.
- Reduce the number of tests by filtering out “uninteresting” genes.

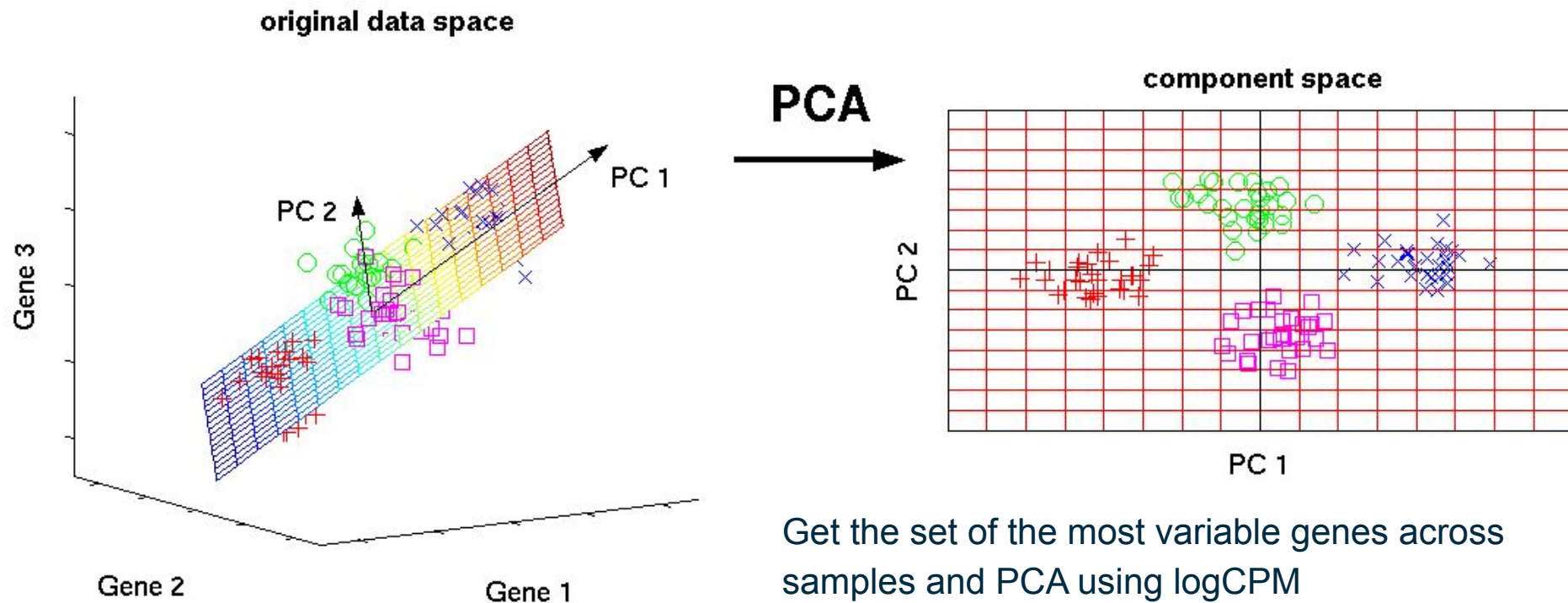
Session 3 Concepts:



Session 3 Demo

Assessing overall similarity across samples

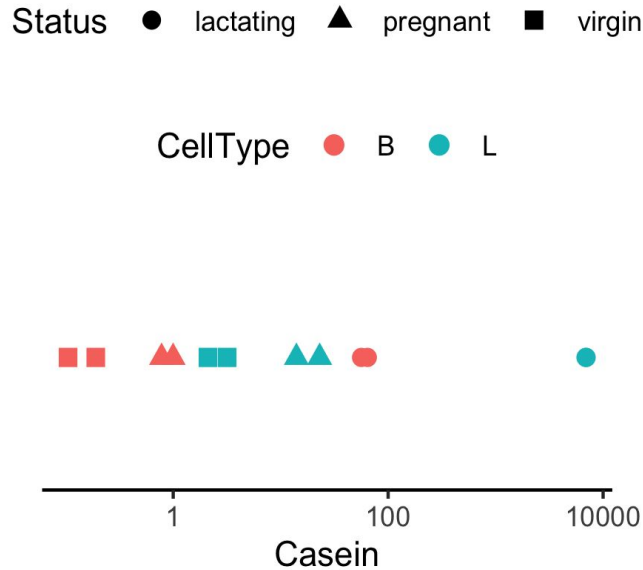
- Which samples are similar to each other, which are different?
- Does this fit to the expectation from the experiment's design?
- What are the major sources of variation in the dataset?



Expression level of Casein varies in a way that is strongly indicative of the effect of CellType and Status

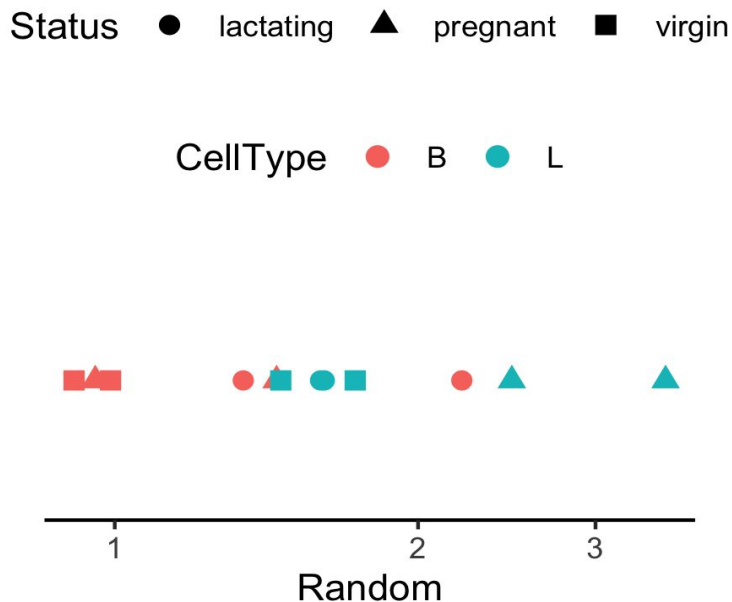
Why are the B.lactating samples not close to B.virgin and B.pregnant samples?

Could it be due to batch effects?



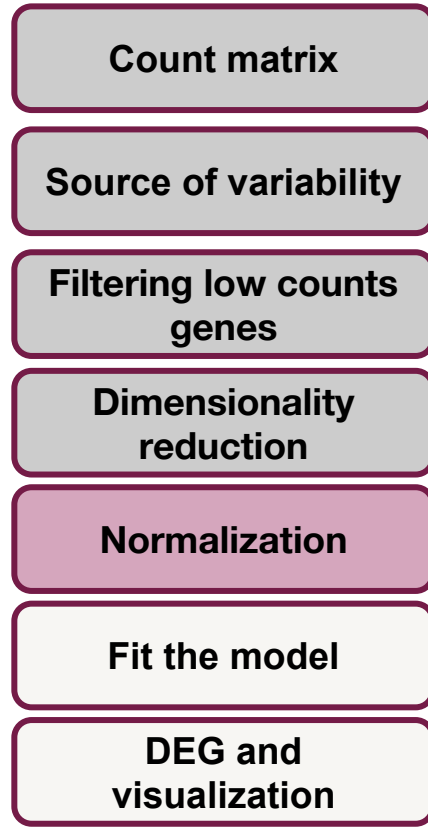
Expression appears to vary across samples but...

In general, the way expression appears to vary across samples could be dominated by noise, batch effects, real signal, etc.



Identify the source for technical variability!

Session 3 Concepts:

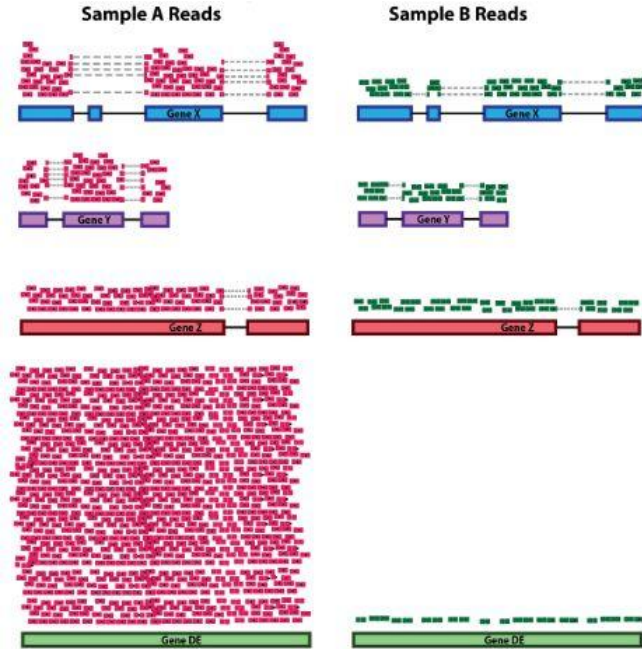


Session 3 Demo

Normalization approach to identify DE genes: edgeR

- *Bioconductor package edgeR*
- *Null model - differences due to noise in counts data*
- If the probability is very low (e.g., $p < 0.05$), infer that something may be happening that we did not account for in the null model. (e.g., biological processes in L cells for milk production) - biological signal

To perform DEG - additional source of bias to account for



Library size 1M

Sample A (where the gene is overexpressed)

Gene DE 500,000 raw counts

All other genes 500,000

Sample B (where the gene is not overexpressed)

Gene DE 10,000 raw counts

All other genes 990,000

If a gene is overexpressed in one group of samples, the other genes look lower expressed

Different RNA sample composition

Cells don't necessarily produce similar levels of RNA/cell between cell types, disease states or developmental stages is not always valid

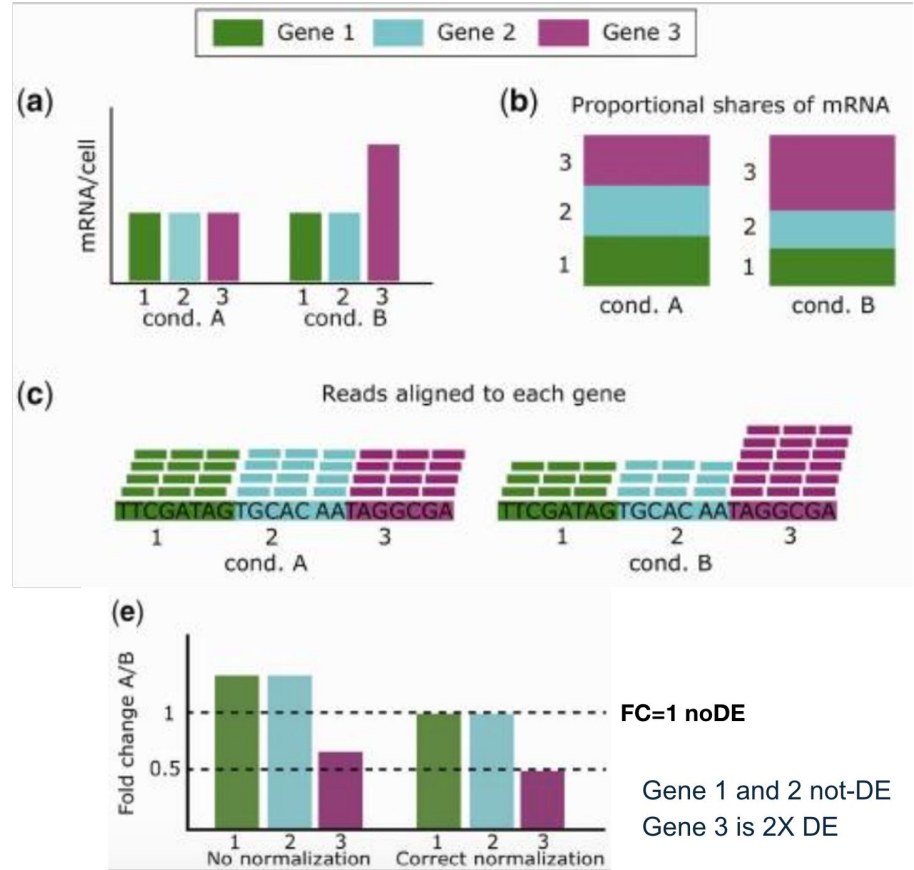
Highly expressed genes bias the real DEG

Total read count for each sample is the same.

Few highly expressed genes make up a greater share of the total molecules (total library size) of samples B.

Smaller fraction of the reads will be left for the other genes for that samples (undersampling).

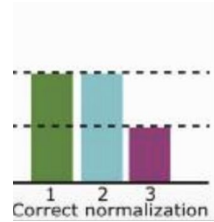
Normalization for DEG: TMM (edgeR), RLE (DESeq)



A few highly differentially expressed genes may have a strong influence on read counts -> minimizing effect of such genes

Assumption: A majority of transcripts is not differentially expressed

Adjust counts such that for most genes, counts are not differential.



Normalize for RNA composition by a **set of scaling factors** that **minimize** the log-fold changes between the samples for most genes.

Approach to identify DE genes: edgeR

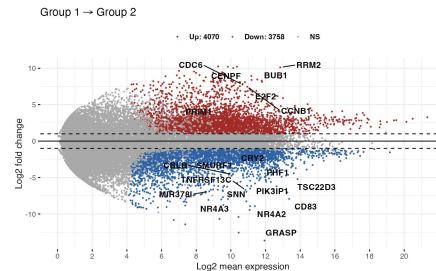
Scaling factors to minimize the log-fold changes between the samples for most genes

- *Reference sample*: have the closest average expressions to the mean of all samples
- *Test samples*: other samples

Generate a gene set removing most/lowest expressed genes (**avg read counts**) and genes with highest/lowest log ratios (**differences in expression**)

Normalization with edgeR: Trimmed Mean of M-values

MA (mean-difference) plot



1. Choose a reference sample.
2. Compute the M and A values for all genes.
M=log FC between ref and test; A=avg count gene between ref and test
3. Filter genes that fall in the tails of M and A distributions.
4. Estimate variance of M values.
5. Estimate TMM --- the weighted average of trimmed M-values.
6. Size factor is 2^{TMM} .
7. Adjust such that these multiply to 1.

lib.size	norm.factors
23218026	1.2368993
21768136	1.2139485
24091588	1.1255640
22656713	1.0698261
21522033	1.0359212
20008326	1.0872153
20384562	1.3684449
21698793	1.3653200
22235847	1.0047431
21982745	0.9232822
24719697	0.5291015
24652963	0.5354877

TMM is implemented in edgeR and performs best for between-samples comparisons, when comparing the samples from different tissues or genotypes.

```
y <- calcNormFactors(y)
```

Other approaches to normalization

Relative Log Expression (RLE) approach by Anders and Huber (2010)

- Reference: geometric mean of all samples
- Normalization factor: median ratio of each sample to the reference
- RLE and TMM give similar robust results with real and simulated data
- *R* package *DESeq* - [tutorial](#)

Upper quartile normalization by Bullard et al (2010)

- Normalization factor: 75% quantile of the counts for each sample
- Not recommended in general - implies a similarity between RNA molecular repertoires expressed

Control genes (housekeeping genes, spike-in) to estimate technical noise ([RUVSeq](#) – 2014 Remove Unwanted Variation)

- powerful when a large set of control genes can be identified

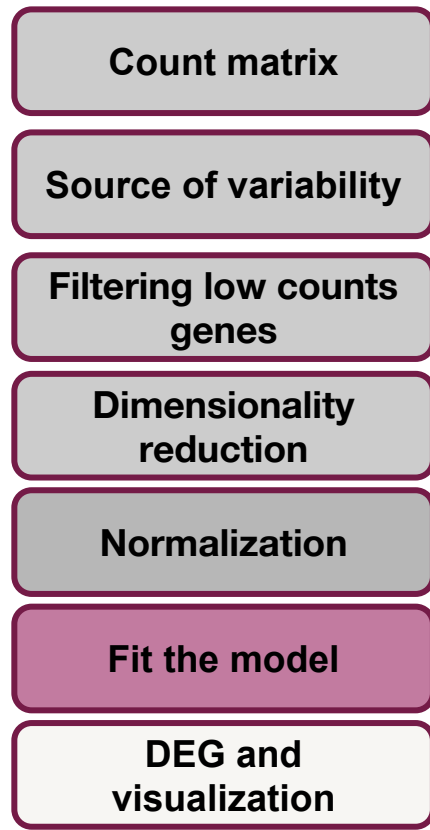
Break (5 min)

Poll 6

What is the purpose of TMM normalization in edgeR?

1. To remove lowly expressed genes
2. To correct for compositional bias between samples
3. To convert counts to $\log_2$ fold changes

Session 3 Concepts:



Session 3 Demo

How to fit the model using the normalized RNA-seq read counts

Model the read mapped to a given gene (Y) and estimate the differences between the two groups/conditions.

$Y \sim \text{group (B.Lactating vs B.Pregnant mice)}$

Total number of reads for a sample $\sim$ millions

Counts per gene $\sim$ tens /hundreds /thousands.

The chance of a *given read* to be mapped to any *specific gene* is rather small.

Discrete events sampled out of a large pool with low probability are usually modeled with Poisson distribution

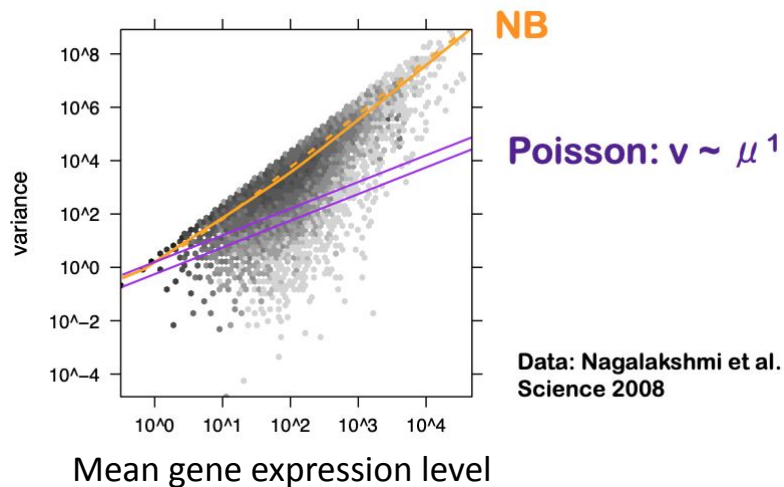
Modeling the read counts mapped to a gene

The number of reads mapped to a gene was first modeled using a **Poisson distribution** (Marioni *et al.* (2008))

Assumption: assumes that mean and variance are the same

BUT the variance grows faster than the mean in RNAseq data.

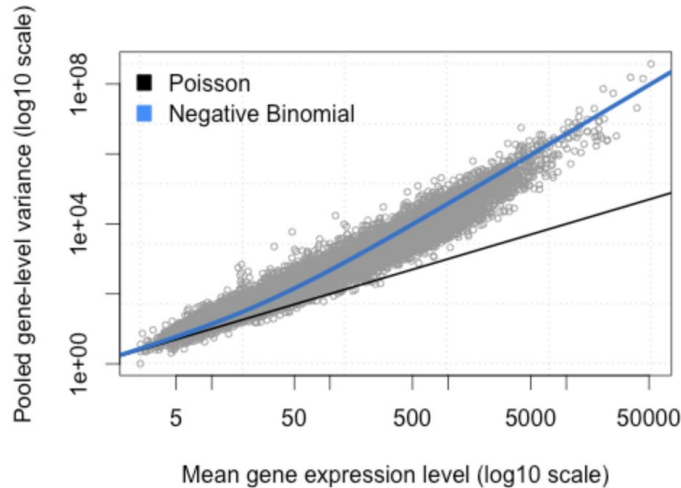
Biological data show more variability than Poisson allows



Why?

Dispersion controls how much biological variability exists between replicates

Underestimation of the biological variance increased the probability to falsely declare a gene DE when it is non-DE (type I error rate)



Negative binomial (NB) distribution

-> alternative to model the read counts for each gene in each sample

The variance is always larger than the mean for the negative binomial $\Rightarrow$ suitable for RNA-seq data

edgeR estimates 3 dispersions

Dispersion ϕ accounts for the variability between biological replicates

- **Common dispersion**: a global dispersion estimate averaged across the genes
- Assumption: every gene has the same biological variability
- Not enough because:
 - counts from biological replicates vary
 - variance is exceeding the mean for highly expressed genes
 - low counts are noisy
 - some genes have stable variance

Genes with similar expression tend to have similar biological variability

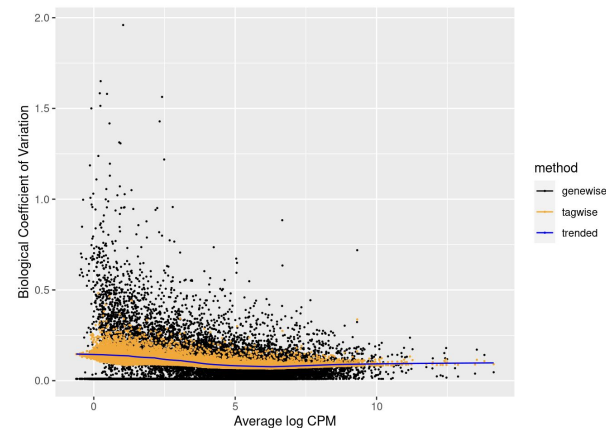
- **Trended dispersion**: dispersion of a gene is predicted from its abundance
 - similar abundant genes
 - Low counts -> high dispersion
 - Higher counts -> lower dispersion

Genes with **similar abundance levels** get similar dispersion estimates

A smooth curve summarizing variability overall

- **Tagwise dispersion**: measure of the degree of consistent inter-library variation for that tag(gene)
- Problem: Gene-specific dispersion estimates are noisy.
- **EB shrinkage** to the trended dispersion - between overall abundance trend and gene specific estimate

Empirical Bayes estimates need to be controlled for the possibility of outlier genes with exceptionally large or small individual dispersions (robust=TRUE)



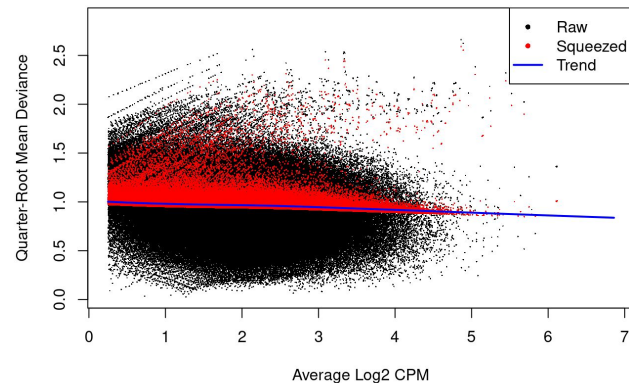
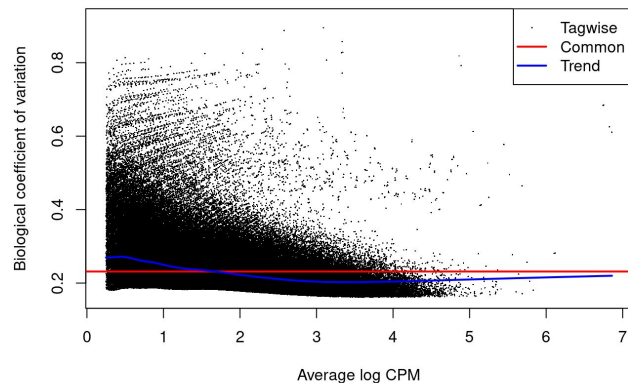
The tag-wise dispersions (orange dots) are the result of shrinking the gene-wise dispersion (black dots) to the trend (blue line)

plotBCV
biological coeff variation
BCV $\approx$ relative variability
between samples

`y <- estimateDisp(y, design)`

Fitting the model : *glmQLFit*

- NB dispersions - higher for genes with very low counts - decrease smoothly with abundance and asymptotically to a constant value for genes with larger counts
- **NB dispersion trend** is used to describe the overall biological variability (fit GLM)
- **EB** shrunk to the trend
- **QL extension** to account for extra gene-specific variability above and below the overall level - reflects how much each gene deviates from the expected variability



edgeR fitting models

QL for bulk RNA-seq:

- + stricter error rate control (more rigorous dispersion and uncertainty)
- + speed improvement compared to other quasi-methods
- + for multiple treatment factors and with small # of biological replicates
- + relative changes in expression levels between conditions (not absolute)

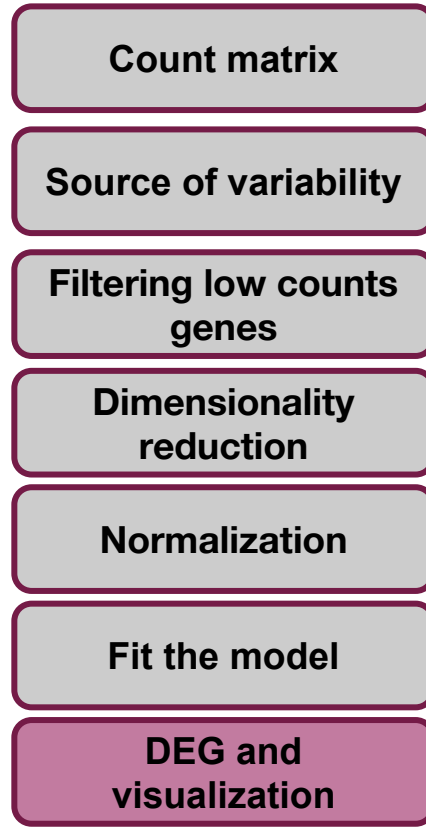
Limma package for large scale datasets – high overlap across methods

Poll 7

Why is RNA-seq count data modeled using a Negative Binomial distribution instead of Poisson?

1. NB is faster than Poisson
2. The variability between biological replicates is larger than expected according to a Poisson model
3. Poisson cannot model fold changes

Session 3 Concepts:



Session 3 Demo

Get the DE genes - *glmQLFTest* and *glmTreat*

- Identifies differential expression based on statistical significance regardless of how small the difference might be -> 5000 DE genes between condition and control groups
- Interested only in genes with large expression changes -> subset of genes more biologically meaningful.
- Modify the statistical test to evaluate variability as well as the magnitude of change of expression values -> expression changes greater than a specified threshold
- Not equivalent to a simple fold change cutoff : “the fold-change below which we are definitely not interested in the gene”
- The total number of DE genes identified at an FDR of 5% can be shown with *decideTest()* – set *cutoff*

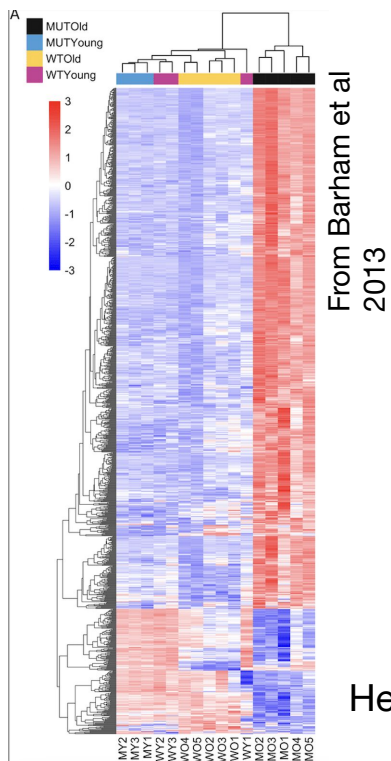
Get the DE genes - *glmQLFTest* and *glmTreat*

	genes	logFC	logCPM	F	PValue	FDR
PCDHA10	PCDHA10	-3.602	5.676	499.9	8.164e-11	1.354e-06
CHGA	CHGA	2.923	5.976	185.4	1.972e-08	0.0001635
ARRB1	ARRB1	-3.914	5.015	158.3	4.627e-08	0.0002019
TSSC2	TSSC2	3.175	3.301	156.8	4.869e-08	0.0002019

Complicated contrasts - *makeContrasts()*

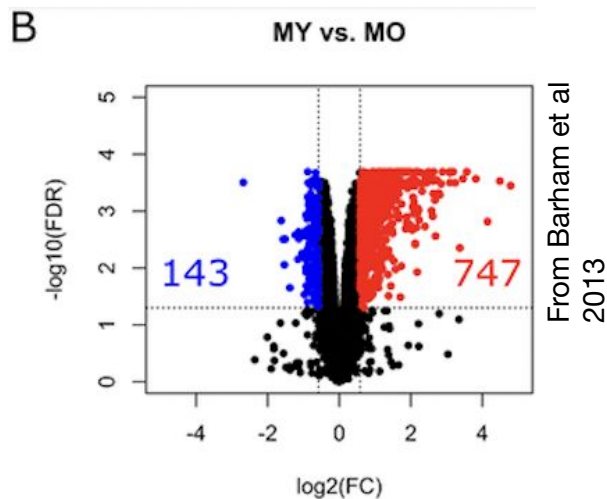
- between lactating and pregnant mice is the same for basal cells as it is for luminal cells
- the interaction effect between mouse status and cell type

Visualization



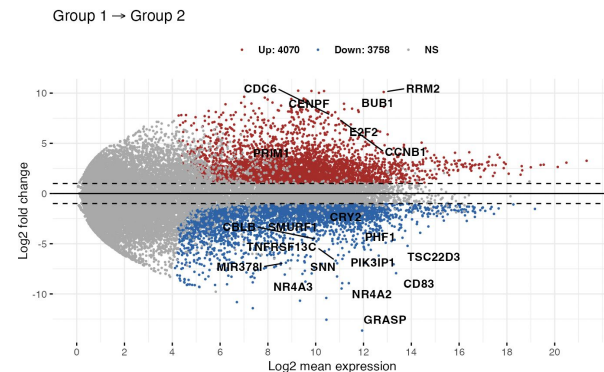
Heatmap

Volcano plot



From Barham et al
2013

MA (mean-difference) plot



Identify genes (and molecular pathways) that are differentially expressed (DE) between two or more biological conditions

Session 3 Concepts:

Count matrix

Source of variability

**Filtering low counts
genes**

**Dimensionality
reduction**

Normalization

Fit the model

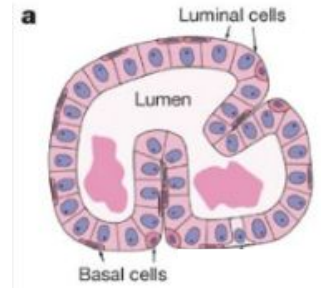
**DEG and
visualization**

Session 3 Demo

Dataset

Transcriptome analysis of luminal and basal cell subpopulations in the lactating versus pregnant mammary gland

- GEO (gene expression omnibus) accession: **GSE60450**
- Tissue of origin: Mammary glands of mouse
- Cell types: Basal stem-cell enriched cells (B) and committed luminal cells (L)
- Biological conditions: Virgin, Lactating (2 day) and Pregnant (18.5 day)
- # of groups: 2 cell types (B/L) x 3 conditions (V/L/P) = 6 groups
- # of replicates: 2 of each group
- Illumina Hiseq sequencer - about 30 million 100bp single-end reads for each sample.



Files for the hands-on session

	GEO	SRA	CellType	Status
MCL1.DG	GSM1480297	SRR1552450	B	virgin
MCL1.DH	GSM1480298	SRR1552451	B	virgin
MCL1.DI	GSM1480299	SRR1552452	B	pregnant
MCL1.DJ	GSM1480300	SRR1552453	B	pregnant
MCL1.DK	GSM1480301	SRR1552454	B	lactating
MCL1.DL	GSM1480302	SRR1552455	B	lactating
MCL1.LA	GSM1480291	SRR1552444	L	virgin

○ [targets.txt](#)

Phenofile

○ [GSE60450_Lactation-GenewiseCounts.txt.gz](#)

Counts for each sample for each gene (Entrez Gene Identifiers)

	Length	MCL1.DG	MCL1.DH	MCL1.DI	MCL1.DJ	MCL1.DK	MCL1.DL	MCL1.LA	MCL1.LB
497097	3634	438	300	65	237	354	287	0	0
100503874	3259	1	0	1	1	0	4	0	0
100038431	1634	0	0	0	0	0	0	0	0
19888	9747	1	1	0	0	0	0	10	3
20671	3130	106	182	82	105	43	82	16	25
27395	4203	309	234	337	300	290	270	560	464

Demo

- o Load and reformat the data
- o Exploratory visualization : MA plot
- o Create DGElist object and retrieve gene symbols
- o Filter genes with inadequate information
- o Normalization
- o Exploratory visualization : MDS and PCA plots
- o Define and fit a model
- o Hypothesis testing (four comparisons)
- o Save results as a table (open in Excel)

Session III

Take-home messages

- Raw counts are not comparable across samples/genes within a sample
- Several normalization methods: some more suitable for DEG
- Estimate the dispersion, visualize the technical variability
- Fit the model: counts variance exceeding the mean
- Make all the comparisons that you wish using complicated contrasts
- Visualize the DEG and get a list for pathway analysis

What if

Q1. Experimental design with repeated measures - alternative methods

Q2. Batch effects in bulk

Real data might need additional analyses choices that need experience

AI - Core position paper

Consult with the Gladstone Bioinformatics core for such scenarios and data.

Upcoming workshops at Gladstone:

AI fluency for Biologists

October 5-6

Single Cell RNA-Seq Analysis using R

October 12-13

Introduction to Pathway Analysis

October 15

Statistics of Enrichment Analysis Methods

October 22-23

Linear Mixed Effects Models Using R

October 26-27

Your feedback is important to us!

Please take the survey ~3 min:

<https://www.surveymonkey.com/r/bioinfo-training26>

The background features a dark teal color with several wavy, overlapping lines in a lighter teal shade. These lines are composed of a grid of small, dashed segments, creating a sense of depth and movement. The lines flow from the left side towards the right, with some peaking and others dipping, giving the impression of a dynamic, fluid environment.

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Additional resources

Helpful resources

- Bacterial reference genome:
 - <https://bacteria.ensembl.org/index.html>
 - STAR and HISAT2 are more commonly used for eukaryotic RNA-seq. For bacterial RNA-seq data, Bowtie2 and BWA can be used as these are non-spliced reads. STAR can also be used for bacterial RNA-seq data but there some additional STAR options that should be specified when doing so.
- How to add reporter sequence to the reference:
 - <https://groups.google.com/g/rna-star/c/FGQRotrCB1Q>
- Cutadapt possible adapters types:
 - <https://cutadapt.readthedocs.io/en/stable/guide.html#adapter-types>

Helpful resources

- Wynton slack channel
 - ucsf-wynton.slack.com
- Gladstone Bioinformatics Core slack channel (email us)
 - <https://gladstoneinstitutes.slack.com/archives/C0145F1L7QS>
- Wynton tutorials
 - <https://github.com/ucsf-wynton/tutorials/wiki>

Helpful resources

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 - <https://groups.google.com/g/rna-star/c/FGQRotrCB1Q>
- Cutadapt possible adapters types:
 - <https://cutadapt.readthedocs.io/en/stable/guide.html#adapter-types>

Tools to manipulate files are available

- Need to sort alignment report?
 - samtools
- Need to convert FASTQ to FASTA?
 - fastx-toolkit
- ...
- Google!

When I need help, which I do need on a daily basis, I visit:

- Slack channel for Wynton users
 - `ucsf-wynton.slack.com`
- <https://stackoverflow.com/>
- Google groups for specific tools
- GitHub issues
- ...

How to publish your code?

Tailored Training Sessions on Publishing Your Code with GitHub

Maximize the impact of your research by mastering GitHub with **Gladstone's Bioinformatics Core training**. The team of experts will help you easily share your code, collaborate with colleagues, publish a function or library, build a website, and more. The core can tailor a training session to meet your lab's specific needs and bring your research to the next level.

Reach out to the core to learn more or set up a training session.

Session III Appendix

How to the expression for each gene within one sample (Sample A)

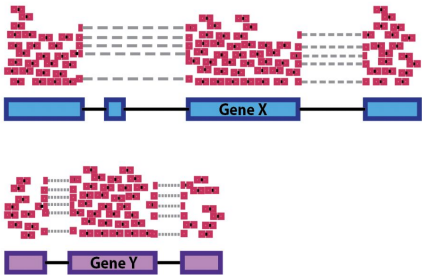
Each column is a sample

Each row is a gene

GENE ID	KD.2	KD.3	OE.1	OE.2	OE.3	IR.1	IR.2	IR.3
1/2-SBSRNA4	57	41	64	55	38	45	31	39
A1BG	71	40	100	81	41	77	58	40
A1BG-AS1	256	177	220	189	107	213	172	126
A1CF	0	1	1	0	0	0	0	0
A2LD1	146	81	138	125	52	91	80	50
A2M	10	9	2	5	2	9	8	4
A2ML1	3	2	6	5	2	2	1	0
A2MP1	0	0	2	1	3	0	2	1
A4GALT	56	37	107	118	65	49	52	37
A4GNT	0	0	0	0	1	0	0	0
AA06	0	0	0	0	0	0	0	0
AAA1	0	0	1	0	0	0	0	0
AAAS	2288	1363	1753	1727	835	1672	1389	1121
AACS	1586	923	951	967	484	938	771	635
AACSP1	1	1	3	0	1	1	1	3
AADAC	0	0	0	0	0	0	0	0
AADACL2	0	0	0	0	0	0	0	0
AADACL3	0	0	0	0	0	0	0	0
AADACL4	0	0	1	1	0	0	0	0
AADAT	856	539	593	576	359	567	521	416
AAGAB	4648	2550	2648	2356	1481	3265	2790	2118
AAK1	2310	1384	1869	1602	980	1675	1614	1108
AAMP	5198	3081	3179	3137	1721	4061	3304	2623
AANAT	7	7	12	12	4	6	2	7
AARS	5570	3323	4782	4580	2473	3953	3339	2666
AARSS	4454	2323	2284	2124	1240	2488	2074	1657

Get the expression for each gene within one sample (Sample A)

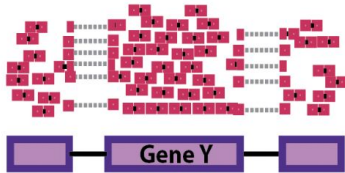
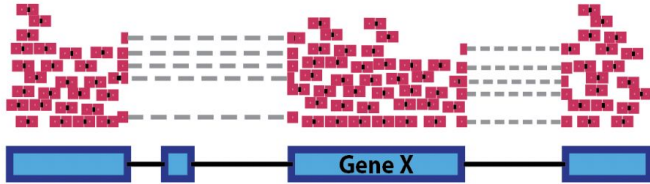
Sample A Reads



Scenario

Which gene is more expressed in Sample A? Gene X or Gene Y?

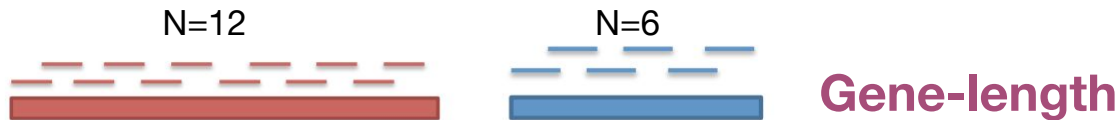
Sample A Reads



What is the source of bias when comparing Gene X and Gene Y within a sample?

Source of bias between genes

At the same expression level, a long gene will have more reads than a shorter gene



Identify and correct technical biases removing the least possible biological signal

- Library size or sequence depth (number of mapped reads)
- Gene length

Commonly used normalization method that includes sequence depth and gene length correction:

- RPKM /FPKM (Reads/Fragments Per Kilobase per Million)
- TPM (Transcripts Per kilobase Million)

RPKM: seq depth and gene length bias

Reads/Fragments Per Kilobase per Million

Gene A 600 bases

Gene B 1100 bases

Gene C 1400 bases

$$\text{RPKM} = 12 / (0.6 * 6) = 3.33$$

$$\text{RPKM} = 24 / (1.1 * 6) = 3.64$$

$$\text{RPKM} = 11 / (1.4 * 6) = 1.31$$



One library with 5 M reads
Total 4 M matched to the
genome sequence
5000 reads matched to a
given gene *with a length of
2000 bp.*



$$\text{RPKM} = 19 / (0.6 * 8) = 3.96$$

$$\text{RPKM} = 28 / (1.1 * 8) = 1.94$$

$$\text{RPKM} = 16 / (1.4 * 8) = 1.43$$

$$\text{RPKM} = \frac{\text{Number of reads mapped to gene} \times 10^3 \times 10^6}{\text{Total number of mapped reads} \times \text{gene length in bp}}$$

$$\text{RPKM} = \frac{5000 \times 10^3 \times 10^6}{4 \times 10^6 \times 2000} = 625$$

RPKM and FPKM -> *the sum of the normalized reads in each sample may be different,*
and this makes it harder to compare samples directly.

TPM: seq depth and gene length bias

TPM: Normalize for gene length first to get the Reads Per Kilobase, sum up all the RPK in a sample (across all genes), and then normalize for sequencing depth

$$\text{TPM} = 10^6 * \frac{\text{reads mapped to transcript} / \text{transcript length}}{\text{Sum}(\text{reads mapped to transcript} / \text{transcript length})}$$

Represent the relative abundance of a transcript among a population of sequenced transcripts

The sum of all TPMs in each samples are the same -> to compare the proportion of reads that mapped to a gene in each sample.

Same denominator -> comparable - NOT for RPKM (different denominator)

Sample1 - gene1 TPM=2.5

Sample2 - gene1 TPM=2.5

Main normalization approaches summary

Normalization method	Description	Accounted factors	Recommendations for use
TPM (transcripts per kilobase million)	counts per length of transcript (kb) per million reads mapped	sequencing depth and gene length	gene count comparisons within a sample or between samples of the same sample group; NOT for DE analysis
RPKM/FPKM (reads/fragments per kilobase of exon per million reads/fragments mapped)	similar to TPM	sequencing depth and gene length	gene count comparisons between genes within a sample; NOT for between sample comparisons or DE analysis

Still not enough to perform DEG analysis

The background features a dark teal color with several wavy, undulating lines in a lighter teal shade. These lines are composed of a fine grid of small dashes or segments, creating a textured, almost 3D effect. The waves flow from the left side towards the right, with some peaks and valleys. The overall aesthetic is modern and scientific.

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